



MR IT
& NETWORKING

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CLASS C - 3 BIT SUBNETTING

STEP BY STEP EXAMPLE



Network Address
192.168.1.0/24
(Class C Network)

Default Subnet Mask
255.255.255.0
(/24)

We borrow **3** bits
from the host portion.

/27

1 BORROW 3 BITS

Default Prefix : /24
Borrowed Bits : 3
New Prefix : /27

New Subnet Mask

255.255.255.224

Binary:

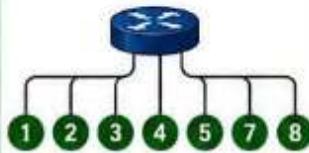
255.255.255.224

11111111.11111111.11111111.11100000

2 NUMBER OF SUBNETS

Formula: 2^n
(n = borrowed bits)

$2^3 = 8$ Subnets



3 HOSTS PER SUBNET

Remaining Host Bits
 $= 8 - 3 = 5$

Formula: $2^n - 2$
(n = host bits)

$2^5 - 2 = 30$ Hosts



30 Usable Hosts
per Subnet

4 BLOCK SIZE (INCREMENT)

Block Size =
256 - New Subnet Mask
(Last Octet)

$256 - 224 = 32$

Subnet Increment = 32



CLASS C /27 SUBNET TABLE (3-BIT SUBNETTING)

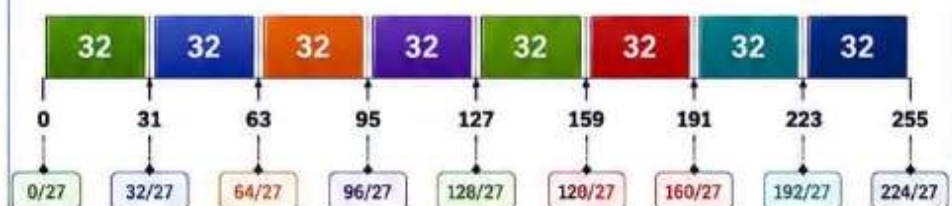
Subnet No.	Network ID (/27)	First Host	Last Host	Broadcast Address	Total Hosts
1	192.168.1.0/27	192.168.1.1	192.168.1.30	192.168.1.31	30
2	192.168.1.32/27	192.168.1.33	192.168.1.62	192.168.1.63	30
3	192.168.1.64/27	192.168.1.65	192.168.1.94	192.168.1.95	30
4	192.168.1.96/27	192.168.1.97	192.168.1.126	192.168.1.127	30
5	192.168.1.128/27	192.168.1.129	192.168.1.158	192.168.1.159	30
6	192.168.1.160/27	192.168.1.161	192.168.1.190	192.168.1.191	30
7	192.168.1.192/27	192.168.1.193	192.168.1.222	192.168.1.223	30
8	192.168.1.224/27	192.168.1.225	192.168.1.254	192.168.1.255	30

QUICK SUMMARY

- ✓ Class : C
- ✓ Default Prefix : /24
- ✓ Borrowed Bits : 3
- ✓ New Prefix : /27
- ✓ Subnet Mask : 255.255.255.224
- ✓ Number of Subnets : 8
- ✓ Hosts per Subnet : 30
- ✓ Block Size (Increment) : 32



VISUAL REPRESENTATION (Last Octet Range)



Each subnet has 30 usable hosts.
Subnet increment is 32 in the last octet.



SUBNET SMARTER. NETWORK STRONGER.
Understand Today, Implement Tomorrow.



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Which protocol is primarily used to automatically assign IP addresses to devices on a network?



A: DNS

B: DHCP

C: FTP

D: SMTP

Cybersecurity For Computer & AI HUB



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COMMENT

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TACACS+ & RADIUS

CONFIGURATION ON CISCO SWITCHES

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1 TACACS+ CONFIGURATION



1. Enable AAA

```
SW1(config)# aaa new-model
```



2. Configure TACACS+ Server

```
SW1(config)# tacacs server TACACS-SRV  
SW1(config-tacacs-server)# address ipv4 192.168.1.10  
SW1(config-tacacs-server)# key cisco123
```



3. Define AAA Authentication

```
SW1(config)# aaa authentication login default group tacacs+ local
```



4. Define AAA Authorization

```
SW1(config)# aaa authorization exec default group tacacs+ local  
SW1(config)# aaa authorization commands 15 default group tacacs+ local
```



5. Define AAA Accounting

```
SW1(config)# aaa accounting exec default start-stop group tacacs+  
SW1(config)# aaa accounting commands 15 default start-stop group tacacs+
```



6. Enable AAA for Console & VTY

```
SW1(config)# line vty 0 15  
SW1(config-line)# login authentication default  
SW1(config-line)# exec authorization default
```



7. (Optional) Fallback to Local

(Local user 'admin' must exist as backup)

```
SW1(config)# aaa authentication login default group tacacs+ local
```

2 RADIUS CONFIGURATION



1. Enable AAA

```
SW1(config)# aaa new-model
```



2. Configure RADIUS Server

```
SW1(config)# radius server RADIUS-SRV  
SW1(config-radius-server)# address ipv4 192.168.1.20 auth-port 1812 acct-port 1813  
SW1(config-radius-server)# key radius123
```



3. Define AAA Authentication

```
SW1(config)# aaa authentication login default group radius local
```



4. Define AAA Authorization

```
SW1(config)# aaa authorization exec default group radius local  
SW1(config)# aaa authorization commands 15 default group radius local
```



5. Define AAA Accounting

```
SW1(config)# aaa accounting exec default start-stop group radius  
SW1(config)# aaa accounting commands 15 default start-stop group radius
```



6. Enable AAA for Console & VTY

```
SW1(config)# line vty 0 15  
SW1(config-line)# login authentication default  
SW1(config-line)# exec authorization default
```



7. (Optional) Fallback to Local

(Local user 'admin' must exist as backup)

```
SW1(config)# aaa authentication login default group radius local
```

VERIFY TACACS+



- SW1# show aaa servers
- SW1# show tacacs-server
- SW1# show aaa authentication
- SW1# show aaa authorization
- SW1# show aaa accounting

VERIFY RADIUS



- SW1# show aaa servers
- SW1# show radius statistics
- SW1# show aaa authentication
- SW1# show aaa authorization
- SW1# show aaa accounting

EXAMPLE LOGIN FLOW



Switch forwards request to server
Server authenticates → Response returned

QUICK NOTES



- TACACS+ supports Authentication, Authorization & Accounting.
- RADIUS supports Authentication & Accounting (Authorization limited).
- Always keep a local user for fallback access.

BEST PRACTICES



- Use strong shared secrets.
- Use dedicated management VLAN.
- Restrict access to AAA servers.
- Enable accounting for better auditing.
- Test failover & fallback scenarios.

IMPORTANT REMINDERS



- Secure your AAA server credentials.
- Use SSH for secure device access.
- Regularly check logs & accounting.
- Keep firmware & configurations updated.



Centralize. Authenticate. Authorize. Account.
Secure your network the smart way!

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**NETWORKING
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CCNA NETWORKING BASIC

BUILDING BLOCKS OF NETWORKING



CCNA Networking covers the fundamentals required to design, build, configure and troubleshoot small to medium networks.

1. NETWORK COMPONENTS



Router

Connects different networks and routes data.



Switch

Connects devices in the same network (LAN).



End Devices

PCs, Servers, Printers, IP Phones that use the network.



Media

Copper (UTP), Fiber Optic, Wireless.

2. OSI MODEL (7 LAYERS)

7



Application

6



Presentation

5



Session

4



Transport

3



Network

2



Data Link

1



Physical

3. TCP/IP MODEL (4 LAYERS)

4

Application Layer

(HTTP, DNS, FTP, SMTP)



3

Transport Layer

(TCP, UDP)



2

Internet Layer

(IP, ICMP)



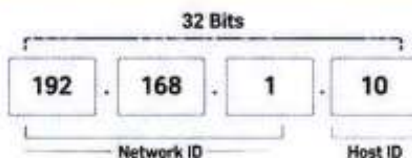
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Network Access Layer

(Ethernet, Wi-Fi)



4. IP ADDRESSING (IPv4)



- ✓ Unique logical address for every device.
- ✓ Made of Network ID + Host ID.
- ✓ Private IP Ranges:

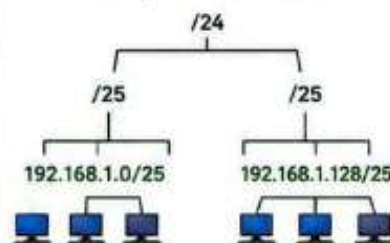
10.0.0.0 – 10.255.255.255

172.16.0.0 – 172.31.255.255

192.168.0.0 – 192.168.255.255

5. SUBNETTING BASIC

Example: 192.168.1.0/24



- ✓ Subnetting divides a network into smaller networks.
- ✓ Helps in efficient IP allocation and reduces broadcast traffic.

6. COMMON PROTOCOLS



ARP

Resolves IP address to MAC address.



DNS

Resolves domain names to IP addresses.



DHCP

Automatically assigns IP addresses.



ICMP

Used for error reporting and network troubleshooting.

7. BASIC NETWORK COMMUNICATION



KEY TAKEAWAY:

Understand these basics strongly to build your networking career.



CCNA is the first step towards becoming a Network Professional!



Learn | Practice | Succeed
Stay Consistent,
Stay Ahead!



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VLAN CONFIGURATION



HR



Account



IT Dept

VLAN 10

- ✓ Switch>enable
- ✓ Switch#configure terminal
- ✓ Switch(config-vlan 10
- ✓ Switch(config-vlan)#name HR
- ✓ Switch(config-vlan)#exit

Assign VLAN to Interface

- ✓ Switch>enable
- ✓ Switch(configure terminal
- ✓ Switch(config-if)#interface gigabitEthernet 0/1
- ✓ Switch(config-if)#switchport mode access
- ✓ Switch(config-if)#switchport access vlan 10
- ✓ Switch(config-if)#exit

VLAN 20

- ✓ Switch>enable
- ✓ Switch#configure terminal
- ✓ Switch(config-vlan 20
- ✓ Switch(config-vlan)#name Account
- ✓ Switch(config-vlan)#exit

Assign VLAN to Interface

- ✓ Switch>enable
- ✓ Switch(configure terminal
- ✓ Switch(config-if)#interface fastEthernet 0/13
- ✓ Switch(config-if)#switchport mode access
- ✓ Switch(config-if)#switchport access vlan 20
- ✓ Switch(config-if)#exit

VLAN 30

- ✓ Switch>enable
- ✓ Switch#configure terminal
- ✓ Switch(config-vlan 30
- ✓ Switch(config-vlan)#name IT
- ✓ Switch(config-vlan)#exit

Assign VLAN to Interface

- ✓ Switch>enable
- ✓ Switch(configure terminal
- ✓ Switch(config-if)#interface fastEthernet 0/25
- ✓ Switch(config-if)#switchport mode access
- ✓ Switch(config-if)#switchport access vlan 30
- ✓ Switch(config-if)#exit

ADVANCED TRUNK PORT

- ✓ Switch(config)#interface gigabitEthernet 1/1
- ✓ Switch(config-if)#switchport mode trunk
- ✓ Switch(config-if)#switchport trunk allowed vlan 10,20,30
- ✓ Switch(config-if)#switchport trunk native vlan 99
- ✓ Switch(config-if)#exit



VOICE & DATA PORT

- ✓ Switch(config)#interface fastEthernet 0/48
- ✓ Switch(config-if)#switchport mode access
- ✓ Switch(config-if)#switchport access vlan 40
- ✓ Switch(config-if)#switchport voice vlan 40
- ✓ Switch(config-if)#exit



PORT SECURITY

- ✓ Switch(config)#interface gigabitEthernet 1/48
- ✓ Switch(config-if)#switchport mode access
- ✓ Switch(config-if)#switchport access vlan 1
- ✓ Switch(config-if)#switchport voice vlan 40
- ✓ Switch(config-if)#exit



PORT SECURITY

- ✓ Switch(config)#interface fastEthernet 0/2
- ✓ Switch(config-if)#switchport port-security
- ✓ Switch(config-if)#switchport port-security mac address sticky
- ✓ Switch(config-if)#switchport port-security violation shutdown
- ✓ Switch(config-if)#exit



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REMEMBER:

- ✓ VLAN = Segmentation
- ✓ Access = End devices
- ✓ Trunk = Carry multiple VLANs
- ✓ Voice VLAN = Separate voice traffic



— VLAN 10 - HR

— VLAN 20 - Account

— VLAN 30 - IT Dept

— VLAN 40 - Voice

— Trunk Link



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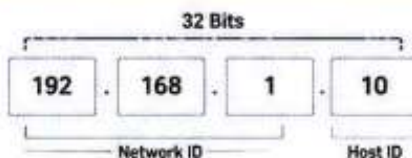
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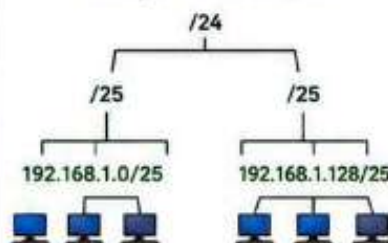
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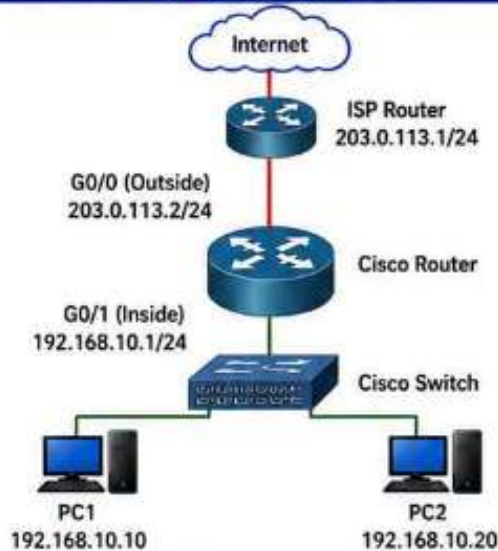
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NAT / PAT CONFIGURATION LAB

Static NAT, Dynamic NAT & PAT Configuration on Cisco Router



1. LAB TOPOLOGY



2. IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask
ISP Router	G0/0	203.0.113.1	255.255.255.0
Cisco Router	G0/0 (Outside)	203.0.113.2	255.255.255.0
Cisco Router	G0/1 (Inside)	192.168.10.1	255.255.255.0
PC1	NIC	192.168.10.10	255.255.255.0
PC2	NIC	192.168.10.20	255.255.255.0

3. CONFIGURE INTERFACES



OUTSIDE

```
interface g0/0
ip address 203.0.113.2 255.255.255.0
ip nat outside
no shutdown
!
```

INSIDE

```
interface g0/1
ip address 192.168.10.1 255.255.255.0
ip nat inside
no shutdown
!
```

4. CONFIGURE PAT (NAT OVERLOAD)



```
access-list 1 permit 192.168.10.0 0.0.0.255

ip nat inside source list 1 interface g0/0 overload
```

The keyword **"overload"** enables PAT (NAT Port Address Translation).

5. CONFIGURE STATIC NAT



```
ip nat inside source static
192.168.10.10 203.0.113.100
```

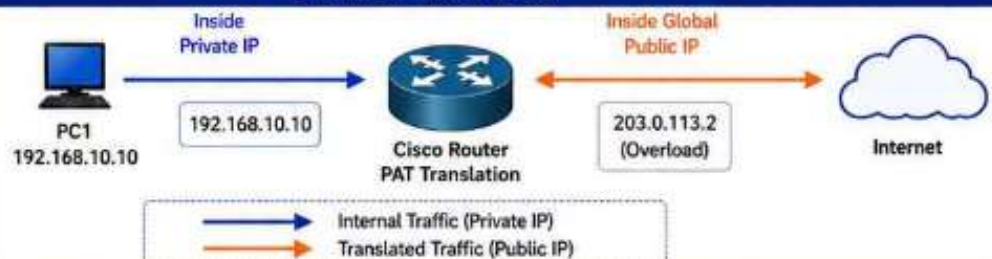
✓ Maps Internal IP (192.168.10.10) to Public IP (203.0.113.100)

6. VERIFICATION COMMANDS



- show ip nat translations
- show ip nat statistics
- show running-config
- show ip interface brief

7. PACKET FLOW (PAT)



8. LAB OBJECTIVES



Configure Inside & Outside Interfaces



Configure PAT (NAT Overload)



Configure Static NAT



Verify NAT Translations



Test Internet Connectivity

Networking With Israr | Learn | Practice | Succeed



CONSTRUCTION SITE CCTV

COMPLETE SECURITY COVERAGE FOR ACTIVE CONSTRUCTION SITES



- 1 Main Entrance Gate LPR Bullet Camera
- 2 Exit Gate Bullet Camera
- 3 Security Guard Booth Dome Camera
- 4 Site Office Dome Camera
- 5 Site Office Entrance Bullet Camera
- 6 Tower Crane PTZ Camera
- 7 Building Front Bullet Camera
- 8 Building Rear Bullet Camera
- 9 Material Storage Yard Dome Camera
- 10 Heavy Equipment Area PTZ Camera
- 11 Concrete Mixing Area Dome Camera
- 12 Worker Assembly Area Dome Camera
- 13 Fuel & Generator Storage Bullet Camera
- 14 Left Perimeter Fence Bullet Camera
- 15 Right Perimeter Fence Bullet Camera
- 16 Rear Access Road Bullet Camera

SYSTEM CONNECTION DIAGRAM



KEY FEATURES



24/7 Recording



AI Human Detection



Motion Detection



Intrusion Detection



PTZ Auto Tracking



Night Vision



4K Ultra HD



Remote Access



IP66 Weatherproof



Construction Site Security



PROTECT WORKERS, EQUIPMENT & MATERIALS WITH A SMART CONSTRUCTION SITE CCTV SYSTEM

IT SUPPORT TROUBLESHOOTING INTERVIEW

1. Computer Not Turning On

Check power cable.
Check power outlet.
Check power button.
Check SMPS / charger.
Remove external devices.
Restart the PC.
If still not working, check hardware and escalate.

2. No Internet Connection

Check network cable / Wi-Fi.
Run ipconfig.
Ping gateway.
Ping Google (8.8.8.8).
Restart network adapter.
Restart router.
Contact network team if needed.

3. Printer Not Printing

Check power.
Check USB / Network cable.
Check paper and toner.
Clear print queue.
Restart Print Spooler.
Reinstall printer driver.

4. Computer Running Slow

Open Task Manager.
Check CPU, RAM, Disk usage.
Close unwanted programs.
Disable startup apps.
Scan for viruses.
Restart computer.
Upgrade RAM / SSD if needed.

5. Forgot Windows Password

Verify user identity.
Reset password in Active Directory.
Ask user to create a new password.
Confirm login works.

6. Blue Screen (BSOD)

Note error code.
Boot into Safe Mode.
Update drivers.
Run sfc /scannow.
Check RAM and hard disk.
Restore Windows if needed.

7. Outlook Not Working

Check internet.
Restart Outlook.
Check mailbox size.
Repair Outlook profile.
Test webmail.
Verify mail server.

8. Shared Folder Access Denied

Check network.
Verify permissions.
Check user group.
Check server availability.
Test with another account.

9. Wi-Fi Not Connecting

Turn Wi-Fi ON.
Restart adapter.
Forget and reconnect network.
Update Wi-Fi driver.
Restart router.
Test another device.

10. Computer Frozen

Wait a few seconds.
Press Ctrl + Shift + Esc.
End frozen application.
Restart PC.
Check Event Viewer.

IMPORTANT COMMANDS

ipconfig	→ Show IP address
ipconfig /all	→ Full network details
ping	→ Check connectivity
tracert	→ Trace network path
nslookup	→ Check DNS
netstat	→ Show network connections
tasklist	→ List running processes
taskkill	→ End process
sfc /scannow	→ Repair system files
DISM	→ Repair Windows image
chkdsk	→ Check disk errors
gpupdate /force	→ Update Group Policy

BASIC TROUBLESHOOTING STEPS

1. Identify the problem.
2. Ask the user questions.
3. Check cables and connections.
4. Test the issue.
5. Apply the solution.
6. Verify the problem is fixed.
7. Document the issue.

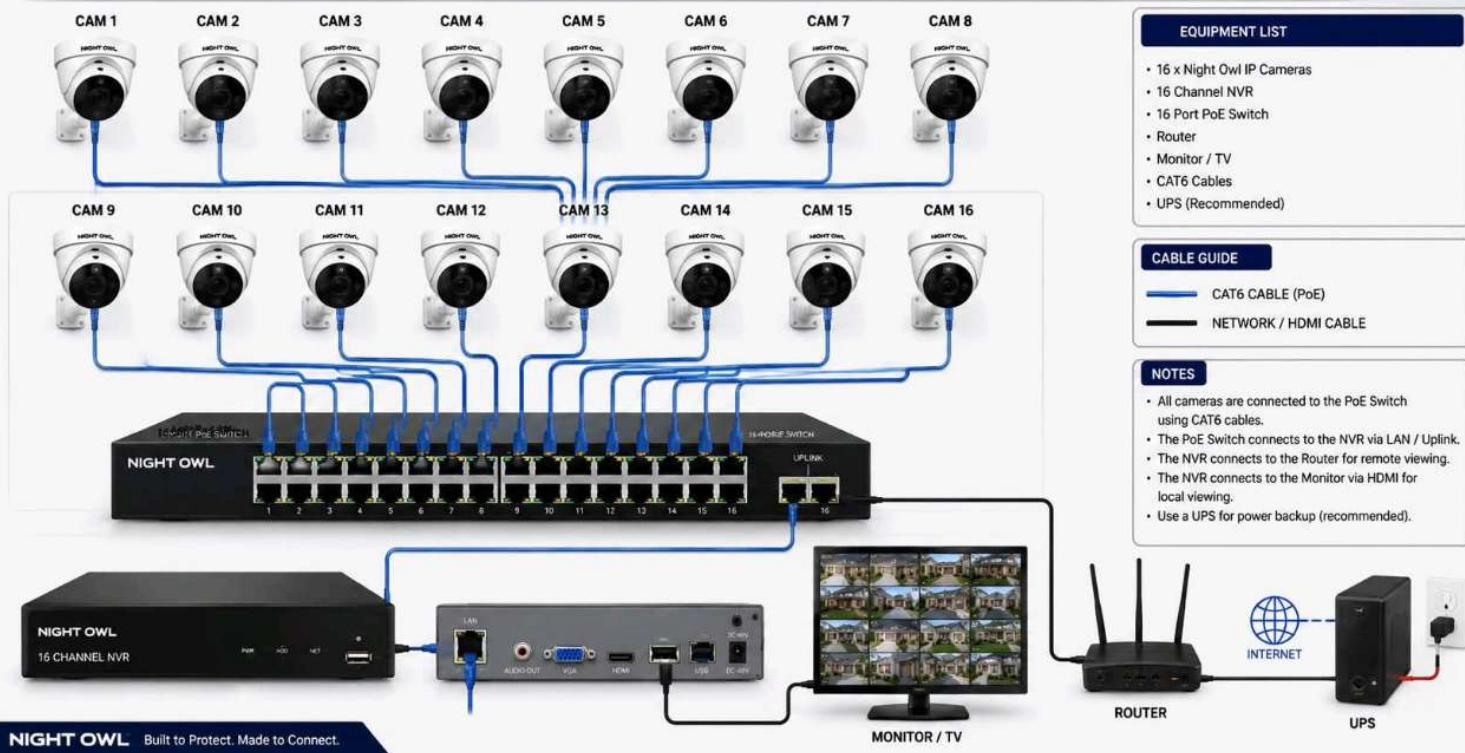
INTERVIEW TIP

Remember this line:

"I always start with the basic checks, identify the root cause, test the solution, verify the issue is resolved, and document the resolution."



INSTALL 16 NIGHTOWL IP CAMERAS



ETHERCHANNEL VS vPC



WHAT IS ETHERCHANNEL?

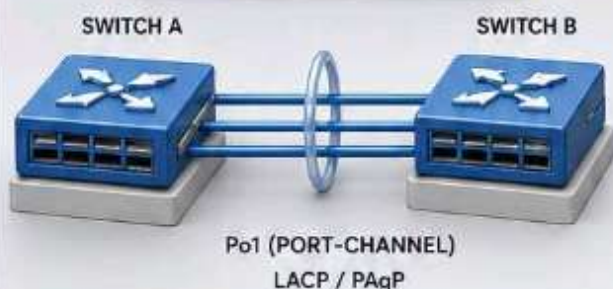
Bundles multiple physical links between two devices into a single logical link to increase bandwidth and provide redundancy.



WHAT IS vPC?

Allows a dual-homed device to connect to two peer switches as if they were a single switch, providing active-active forwarding and high availability.

ETHERCHANNEL TOPOLOGY



vPC TOPOLOGY



KEY DIFFERENCES

FEATURE	ETHERCHANNEL	vPC
PURPOSE	Aggregates links between two devices.	Provides active-active connectivity to two peer switches.
DEVICES INVOLVED	Two devices (Point-to-Point)	Two switches (Peers) + one dual-homed device
LAYERS	Layer 2	Layer 2
STANDARDS / PROTOCOLS	LACP (802.3ad) / PAgP (Cisco) (Static also supported)	Proprietary Cisco Technology
LINK AGGREGATION	Yes – Bundle links into Port-Channel	Yes – Uses Port-Channel between dual-homed device and peers
REDUNDANCY	Active/Standby (Blocked links exist) or Active/Active (with LACP)	Active/Active
USE CASE	Increase bandwidth and redundancy between two switches/routers.	High availability for servers or devices connected to two switches.
SPLIT BRAIN RISK	No	Yes – Requires consistency checks to prevent loops
CONFIGURATION	Simple	Complex (Domain, Peer-Keepalive, Consistency Check, Role Priority)
FAILURE IMPACT	One link down → Traffic uses remaining links	Peer link down → One switch becomes secondary (or dual-active with risk)

BENEFITS OF ETHERCHANNEL



- Increases bandwidth
- Provides link redundancy
- Load balancing across links
- Improves reliability of network links
- Simple to configure and deploy



BENEFITS OF vPC



- Active-active forwarding
- No Spanning Tree blocking on dual-homed links
- Better bandwidth utilization
- High availability for critical applications
- Seamless failover between peer switches



- Use EtherChannel when connecting two devices.
- Use vPC when connecting a device to two switches for high availability.



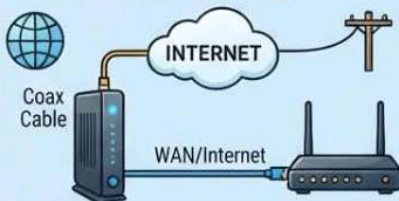
PRO TIP

vPC requires careful design and configuration. Ensure proper peer link, keepalive, and consistency check configurations.



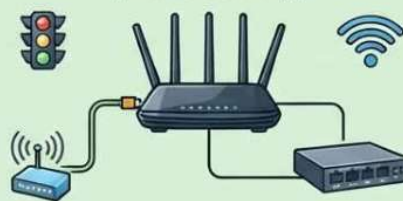
ROUTER vs. MODEM vs. SWITCH: A COMPLETE COMPARISON

MODEM (The Gateway)



- **FUNCTION:** Translates ISP signal (e.g., Cable/DSL) into digital data.
- **CONNECTION:** Connects local network to the WAN (Internet).
- **ONE-TO-ONE:** typically one public IP address.
- **OPERATES AT:** Layer 1 (Physical)

ROUTER (The Traffic Cop)



- **FUNCTION:** Directs data packets between different networks (e.g., Home LAN to Internet).
- **CONNECTION:** Creates and manages a local network (LAN).
- **SECURITY:** Includes Firewall, NAT, DHCP.
- **OPERATES AT:** Layer 3 (Network)

SWITCH (The Data Distributor)



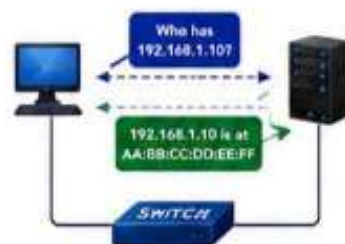
- **FUNCTION:** Connects devices within the same local network (LAN).
- **CONNECTION:** Expands the number of available Ethernet ports.
- **EFFICIENCY:** Uses MAC addresses to forward data to specific ports.
- **OPERATES AT:** Layer 2 (Data Link)

Feature	MODEM	ROUTER	SWITCH
Primary Role	Translates ISP signal	Directs data packets	Connects devices
Data Handling	Legal, digital data.	Data Handling data	Local Handling data
Local Network	Local Network	Creates Network	Internet devices
Internet Access	Only and public	Manages local manages	Volunteers to local addresses
Port Count	6	84	16

ARP PROTOCOL

ADDRESS RESOLUTION PROTOCOL

ARP (Address Resolution Protocol) is used to find the MAC address of a device when we know its IP address. It works at Layer 2 of the OSI model and is essential for communication in IPv4 networks.



1 WHAT IS ARP?

ARP maps an IPv4 address to a MAC (Hardware) address.

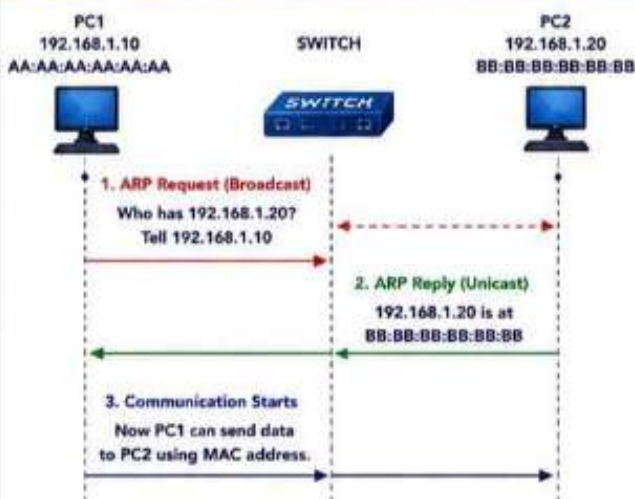


Input : IP Address
Output : MAC Address



Purpose:
Allows devices in the same network to communicate with each other.

2 HOW ARP WORKS (EXAMPLE)



3 ARP PACKET FORMAT

Field	Size (Bytes)
Hardware Type	2
Protocol Type	2
Hardware Size	1
Protocol Size	1
Opcode	2
Sender MAC Address	6
Sender IP Address	4
Target MAC Address	6
Target IP Address	4

Opcode Values
• 1 = ARP Request
• 2 = ARP Reply

4 ARP PROCESS (DETAILED EXAMPLE)



STEP 1: ARP REQUEST (BROADCAST)

PC1 wants to talk to PC2 (192.168.1.20) but does not know its MAC address.

- PC1 sends a broadcast ARP request.
- Source IP: 192.168.1.10
- Source MAC: AA:AA:AA:AA:AA:AA
- Target IP: 192.168.1.20
- Target MAC: FF:FF:FF:FF:FF:FF (Broadcast)

Who has 192.168.1.20? Tell 192.168.1.10

STEP 2: ARP REPLY (UNICAST)

PC2 receives the request and replies back to PC1.

- Source IP: 192.168.1.20
- Source MAC: BB:BB:BB:BB:BB:BB
- Target IP: 192.168.1.10
- Target MAC: AA:AA:AA:AA:AA:AA (Unicast)

192.168.1.20 is at BB:BB:BB:BB:BB:BB

STEP 3: COMMUNICATION

PC1 now knows PC2's MAC address and can send frames directly.

5 TYPES OF ARP



1. Dynamic ARP

Standard ARP process. Entries are added automatically and have a limited lifetime.



2. Static ARP

Manually configured ARP entries. They do not age out.



3. Gratuitous ARP

A device sends an ARP reply for its own IP address to announce or update its presence.

6 ARP CACHE (ARP TABLE)

Devices store ARP entries in an ARP cache (table).

IP Address	MAC Address	Type	Age (sec)
192.168.1.10	AA:AA:AA:AA:AA:AA	Dynamic	120
192.168.1.20	BB:BB:BB:BB:BB:BB	Dynamic	115
...	...	...	...

Use command: `arp -a` (Windows) `show arp` (Cisco)

7 ARP USES

- ✓ Find MAC address of a device in the local network.
- ✓ Required for sending data to another device.
- ✓ Used by default gateway when sending packets outside local network.
- ✓ Essential for IPv4 communication.



8 IMPORTANT NOTES

- ✓ ARP works only in the local broadcast domain.
- ✓ ARP does not work across routers.
- ✓ ARP is an IPv4 protocol.
- ✓ Uses broadcast for request and unicast for reply.
- ✓ Vulnerable to ARP Spoofing / ARP Poisoning (Man-in-the-Middle attacks).



IN SHORT

ARP helps devices find MAC addresses using IP addresses so they can communicate at Layer 2 (Data Link Layer).



TRUNKING LAB

CONFIGURE AND VERIFY 802.1Q TRUNKING BETWEEN SWITCHES

OBJECTIVE

Configure 802.1Q trunk between switches to allow multiple VLANs to pass through a single link.

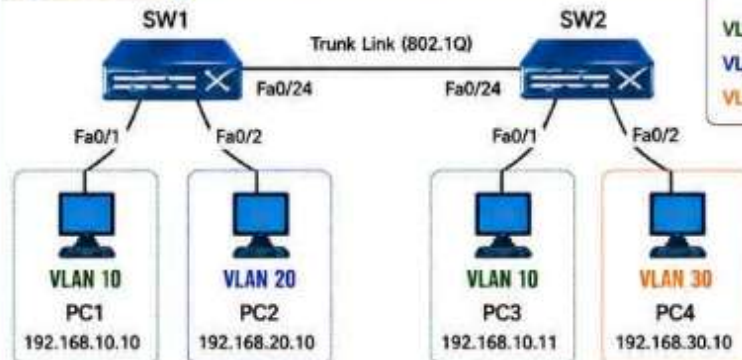


TASKS

- 1 Create VLANs 10, 20, 30 on both switches.
- 2 Configure access ports for each VLAN.
- 3 Configure the link between switches as an 802.1Q trunk.
- 4 Allow VLANs 10, 20, 30 on the trunk.
- 5 Verify trunk status.
- 6 Test connectivity between PCs in different switches within the same VLAN.



TOPOLOGY



VLAN INFORMATION

VLAN 10 → 192.168.10.0 /24
VLAN 20 → 192.168.20.0 /24
VLAN 30 → 192.168.30.0 /24

PORT ASSIGNMENT TABLE

Switch	Port	Mode	Assigned VLAN	Description
SW1	Fa0/1	Access	10	PC1
SW1	Fa0/2	Access	20	PC2
SW1	Fa0/24	Trunk	10,20,30	To SW2
SW2	Fa0/1	Access	10	PC3
SW2	Fa0/2	Access	30	PC4
SW2	Fa0/24	Trunk	10,20,30	To SW1

VERIFICATION CHECKS

- ✓ Trunk link is up.
- ✓ VLANs 10, 20, 30 are allowed on the trunk.
- ✓ Access ports are in the correct VLANs.
- ✓ PCs in the same VLAN (across switches) can ping each other.



SW1 - CONFIGURATION

```
1 enable
2 configure terminal
3 vlan 10
4 name SALES
5 vlan 20
6 name HR
7 vlan 30
8 name FINANCE
9 interface range fa0/1 - 2
10 switchport mode access
11 switchport access vlan 10
12 interface fa0/1
13 switchport access vlan 10
14 interface fa0/2
15 switchport access vlan 20
16 interface fa0/24
17 switchport mode trunk
18 switchport trunk allowed vlan 10,20,30
19 end
20 write memory
```



SW2 - CONFIGURATION

```
1 enable
2 configure terminal
3 vlan 10
4 name SALES
5 vlan 20
6 name HR
7 vlan 30
8 name FINANCE
9 interface range fa0/1 - 2
10 switchport mode access
11 interface fa0/1
12 switchport access vlan 10
13 interface fa0/2
14 switchport access vlan 30
15 interface fa0/24
16 switchport mode trunk
17 switchport trunk allowed vlan 10,20,30
18 end
19 write memory
```



VERIFICATION COMMANDS

- show vlan brief
- show interfaces trunk
- show interfaces fa0/24 switchport
- show running-config
- ping 192.168.10.11 (from PC1)
- ping 192.168.30.10 (from PC1)
- ping 192.168.10.10 (from PC3)
- ping 192.168.20.10 (from PC3)



These commands help you verify trunk status, allowed VLANs and end-to-end connectivity.

TRUNK & ACCESS QUICK RECAP

ACCESS PORT	TRUNK PORT
Belongs to a single VLAN	Carries multiple VLANs
Used to connect end devices (PCs, Phones, APs)	Used to connect switches or routers
Sends/receives untagged frames (for that VLAN)	Sends/receives tagged frames (802.1Q)
Fa0/1 (VLAN 10)	Fa0/24 (Trunk 10,20,30)

TESTING CONNECTIVITY MATRIX

Source PC	Destination PC	Same VLAN?	Expected Result
PC1 (VLAN 10)	PC3 (VLAN 10)	Yes	Ping Success
PC1 (VLAN 10)	PC2 (VLAN 20)	No	Ping Fail
PC2 (VLAN 20)	PC4 (VLAN 30)	No	Ping Fail
PC3 (VLAN 10)	PC1 (VLAN 10)	Yes	Ping Success
PC4 (VLAN 30)	PC3 (VLAN 10)	No	Ping Fail
PC4 (VLAN 30)	PC4 itself	Yes	Ping Success

CHALLENGE QUESTION



If VLAN 20 is not allowed on the trunk link between SW1 and SW2, what will happen when PC2 (VLAN 20) tries to ping PC3 (VLAN 20) after moving PC3 to VLAN 20?

Why?

How will you fix it?

Comment your answer below!

QUICK TIPS

✓ Always use consistent VLAN IDs and names.

✓ Verify trunk allowed VLANs after configuration.

✓ Use show commands to troubleshoot.

✓ Test connectivity from both sides.

✓ Save your configuration regularly.





NETWORKING
WITH ISRAR

LEARN | PRACTICE | SUCCEED

BANDWIDTH

VS

THROUGHPUT vs LATENCY



Three important network performance concepts that are often confused, but mean very different things.

BANDWIDTH



The maximum amount of data that can be transmitted over a network connection in a given time.

Think of it as:
The size of the pipe



Measured in:

bps (bit per second)
Kbps, Mbps, Gbps



Example:

Your internet plan is 100 Mbps bandwidth. That's the max capacity your connection can handle.

THROUGHPUT



The actual amount of data successfully transferred over a network connection in a given time.

Think of it as:
The water flowing in the pipe



Measured in:

bps (bit per second)
Kbps, Mbps, Gbps



Example:

Your 100 Mbps connection may give you 60 Mbps throughput due to network conditions, congestion, etc.

LATENCY



The time it takes for a small piece of data to travel from the source to the destination and back.

Think of it as:
The time it takes for the first drop of water to reach you



Measured in:

Milliseconds (ms)
Lower is better



Example:

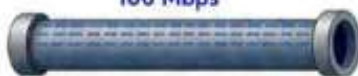
A latency of 20 ms is good (fast response). A latency of 200 ms is high (noticeable delay).

REAL-WORLD EXAMPLE



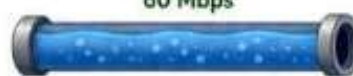
INTERNET

BANDWIDTH
100 Mbps



NETWORK

THROUGHPUT
60 Mbps



YOUR DEVICE



LATENCY
20 ms



PM Networking

Konten AI · 2 hr · 🌐

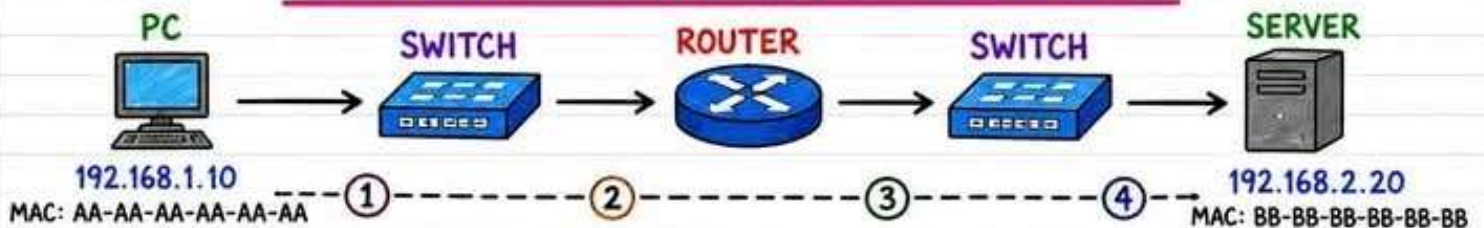


Packet Flow

Lihat terjemahan

⌞ PACKET FLOW ⌞

PC → SWITCH → ROUTER → SWITCH → SERVER



★ HOW THE PACKET FLOW HAPPENS ★

① PC → SWITCH

- ★ PC checks the destination IP (192.168.2.20)
- ★ It sees the destination is in another network.
- ★ So, it sends the packet to Default Gateway (Router).
- ★ It uses ARP to find the Router's MAC.
- ★ Frame sent to SWITCH.

② SWITCH → ROUTER

- ★ Switch reads the destination MAC (Router's MAC).
- ★ It forwards the frame to the Router.

③ ROUTER → SWITCH

- ★ Router removes Layer 2 header (decapsulation).
- ★ Checks the destination IP (192.168.2.20) and finds the best path.
- ★ Router does ARP for Server's MAC.
- ★ Creates a new frame with new MAC addresses and forwards to SWITCH.

④ SWITCH → SERVER

- ★ Switch reads the destination MAC (Server's MAC).
- ★ It forwards the frame to the Server.
- ★ Server receives the packet. ✓

IMPORTANT

• WHAT REMAINS SAME (End to End)
IP Addresses
(Source IP & Destination IP) 🔒

• WHAT CHANGES AT EVERY HOP
MAC Addresses
(Source MAC & Destination MAC) ↻

PM NETWORKING ❤️

WHAT IS VXLAN?

VXLAN

VXLAN (Virtual Extensible LAN) is an overlay network technology that extends Layer 2 networks over Layer 3 IP networks.

It solves the scalability limitations of VLAN by using a 24-bit segment ID, allowing up to 16 million virtual networks.

VXLAN



VXLAN PACKET FORMAT



KEY FEATURES

- ✓ Uses UDP 4789 for transport.
- ✓ Supports up to 16 million segments (24-bit VXLAN ID).
- ✓ Extends Layer 2 networks over Layer 3 infrastructure.
- ✓ Decouples overlay network from underlying physical network.
- ✓ Enables multi-tenancy and network virtualization.
- ✓ Works across data centers and cloud environments.

KEY COMPONENTS

- VTEP (VXLAN Tunnel End Point)**
Encapsulates and decapsulates VXLAN packets.
- VXLAN Segment (VNI)**
Logical Layer 2 network identified by a 24-bit ID.
- Control Plane**
Used to learn MAC addresses and VTEP information (e.g., BGP EVPN).
- Underlay Network**
IP-based network that transports VXLAN traffic.

BENEFITS OF VXLAN



IN SHORT:

VXLAN enables efficient, scalable, and flexible Layer 2 extension over Layer 3 networks using encapsulation and tunneling.



www.thenetworkdna.com

PUBLIC IP

Connects your network to the Internet

Example: 203.0.113.25



- 🌐 Used to identify your network on the Internet
- 📍 Assigned by your ISP
- 👤 Globally unique
- 🔒 Reachable from anywhere on the Internet
- 💰 May change (or be static with extra cost)

VS

PRIVATE IP

Used within your local network

Example: 192.168.1.10



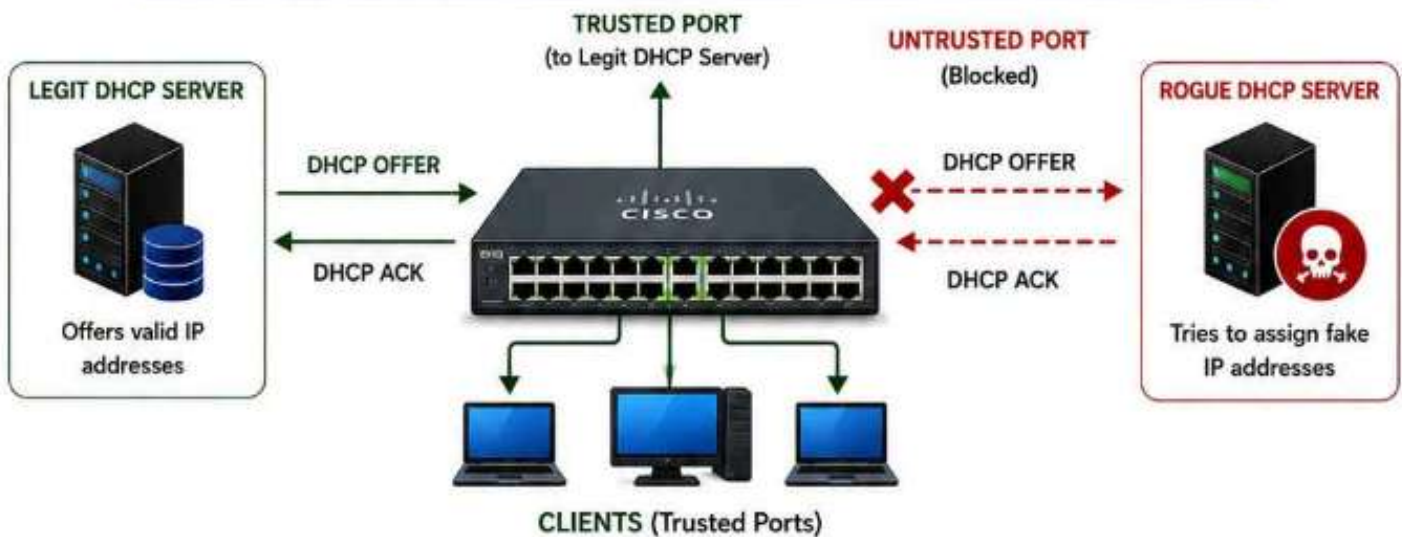
- 🏠 Used to identify devices within a local network
- 📶 Assigned by your router (DHCP)
- 👥 Not unique, can be reused in different networks
- 🛡️ Not reachable directly from the Internet
- 💰 Free

DHCP SNOOPING

Stop Rogue DHCP Attacks



DHCP Snooping is a Layer 2 security feature that **prevents unauthorized DHCP servers** from assigning IP addresses on your network. It builds a trusted database of valid DHCP responses and blocks **malicious DHCP** messages.



HOW DHCP SNOOPING WORKS



1
Monitors all DHCP messages on all ports.



2
Builds a database of trusted DHCP bindings.



3
Validates DHCP messages against the database.



4
Blocks invalid DHCP messages from unauthorized servers.

KEY BENEFITS

- ✓ Prevents rogue DHCP servers
- ✓ Protects against IP address theft
- ✓ Stops man-in-the-middle attacks
- ✓ Ensures only trusted DHCP responses
- ✓ Improves overall network security



BEST PRACTICES

- Trust only the port connected to the legitimate DHCP server.
- Verify DHCP bindings regularly.
- Enable DHCP Snooping on all switches.
- Use with other security features like IP Source Guard and DAI.

3 GALAKSI YANG PALING BERDEKATAN

Tiga galaksi spiral besar yang berada dalam Grup Lokal (Local Group),
dan memiliki jarak relatif dekat dengan Bumi.

BIMA SAKTI
(Milky Way)

ANDROMEDA
(M31)

TATA SURYA

JARAK BUMI KE
ANDROMEDA
±2,54 JUTA
TAHUN CAHAYA

TRIANGULUM
(M33)

JARAK BUMI KE
TRIANGULUM
±2,73 JUTA
TAHUN CAHAYA



TATA SURYA

Lokasi kita di galaksi Bima Sakti,
berada di Lengan Orion,
sekitar 26.000 tahun cahaya
dari pusat galaksi.



BIMA SAKTI (Milky Way)

- Tipe : Spiral Batang
- Diameter : ±100.000 tahun cahaya
- Anggota : ±100 - 400 miliar bintang



ANDROMEDA (M31)

- Tipe : Spiral
- Diameter : ±220.000 tahun cahaya
- Anggota : ±1 triliun bintang



TRIANGULUM (M33)

- Tipe : Spiral
- Diameter : ±60.000 tahun cahaya
- Anggota : ±40 miliar bintang



Ketiga galaksi ini bersama puluhan galaksi kecil lainnya membentuk Grup Lokal (Local Group),
dan diperkirakan akan saling berinteraksi di masa depan.

One Neighbor Is Silent— How Wide Is the Problem?

One Silent OSPF Neighbor



Scope →
Fault Domain Troubleshooting

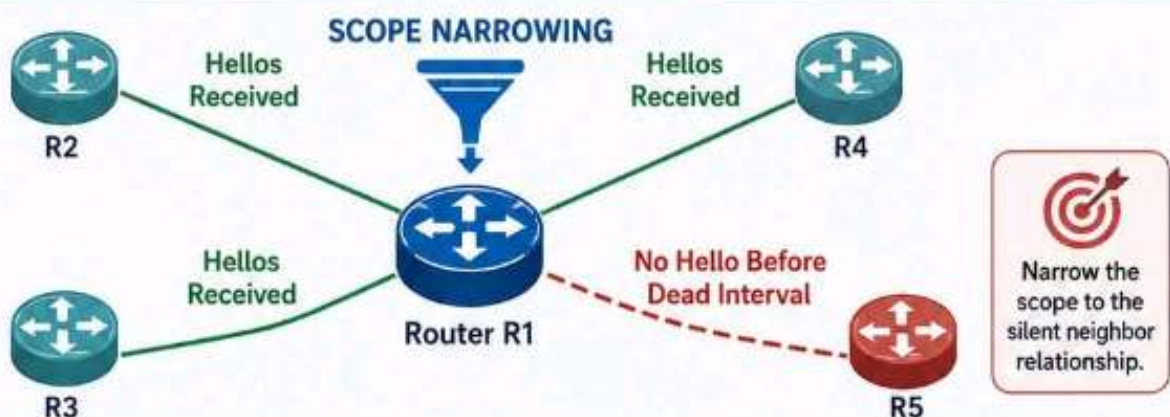


Difficulty: Beginner



SCENARIO

A router keeps receiving Hello messages from several OSPF neighbors, but one neighbor stays silent for the entire Dead Interval. What does this affected-versus-unaffected comparison suggest about the likely scope of the issue?



WHAT THE COMPARISON SUGGESTS

The issue may be limited to the neighbor relationship involving the silent router. Because the other OSPF neighbor relationships continue receiving Hellos normally, a router-wide or OSPF-wide failure is less likely.



WHY THIS MATTERS

- ✓ OSPF expects periodic Hello messages from each neighbor.
- ✓ One neighbor did not send a Hello before the Dead Interval expired.
- ✓ Other neighbor relationships are still active and receiving Hellos.
- ✓ This helps narrow the likely problem area to the affected neighbor relationship.
- ✓ It does not prove the exact reason for the silence.

WORKS VS FAILS



STILL WORKING

- Several neighbor Hellos continue arriving
- Other neighbor relationships remain active



AFFECTED

- One neighbor stays silent
- Dead Interval expires for that relationship



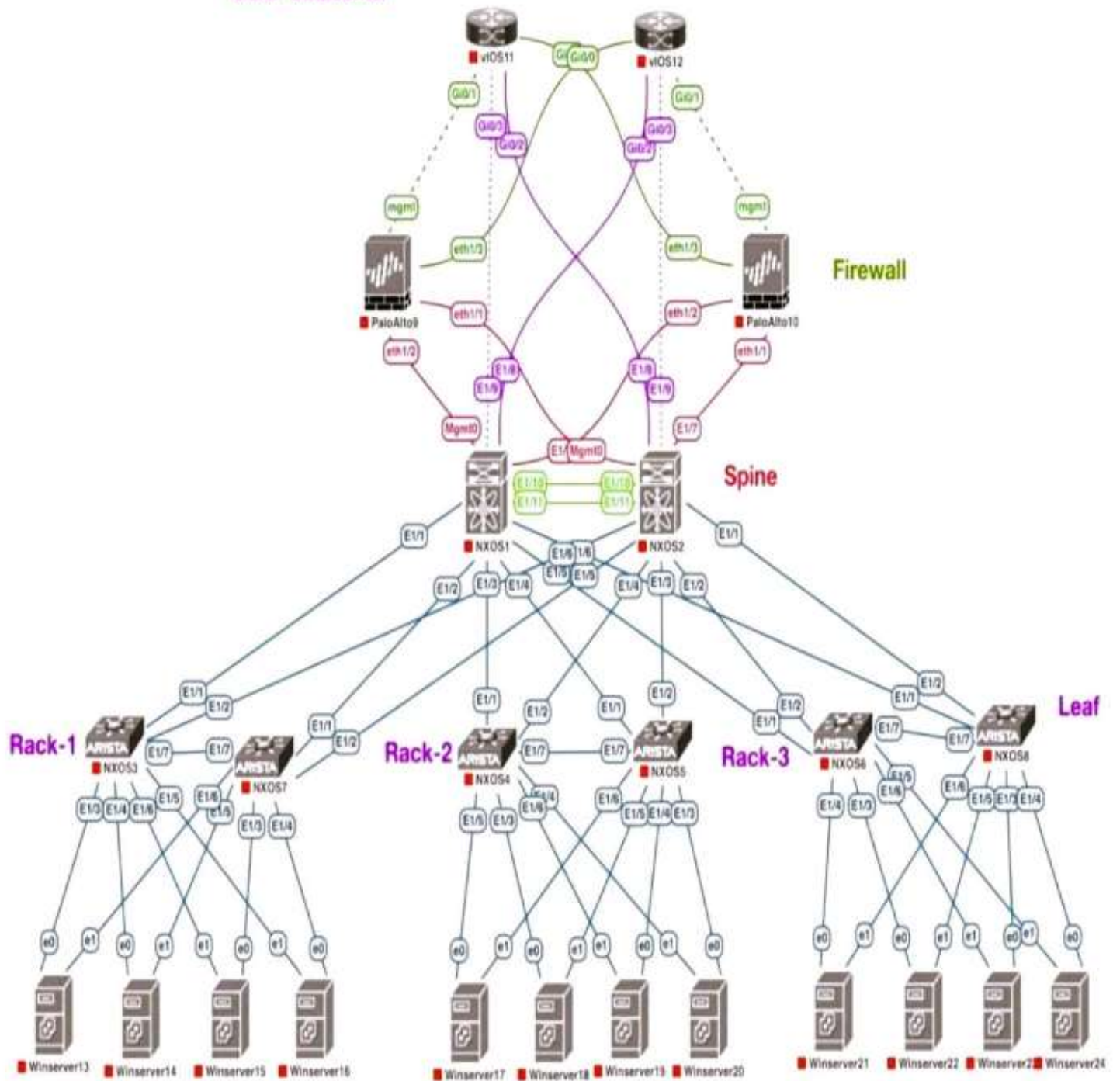
KEY TAKEAWAY

When most OSPF neighbor relationships still receive Hellos, the fault domain is more likely narrowed to the silent neighbor relationship rather than the entire router or the whole OSPF process.



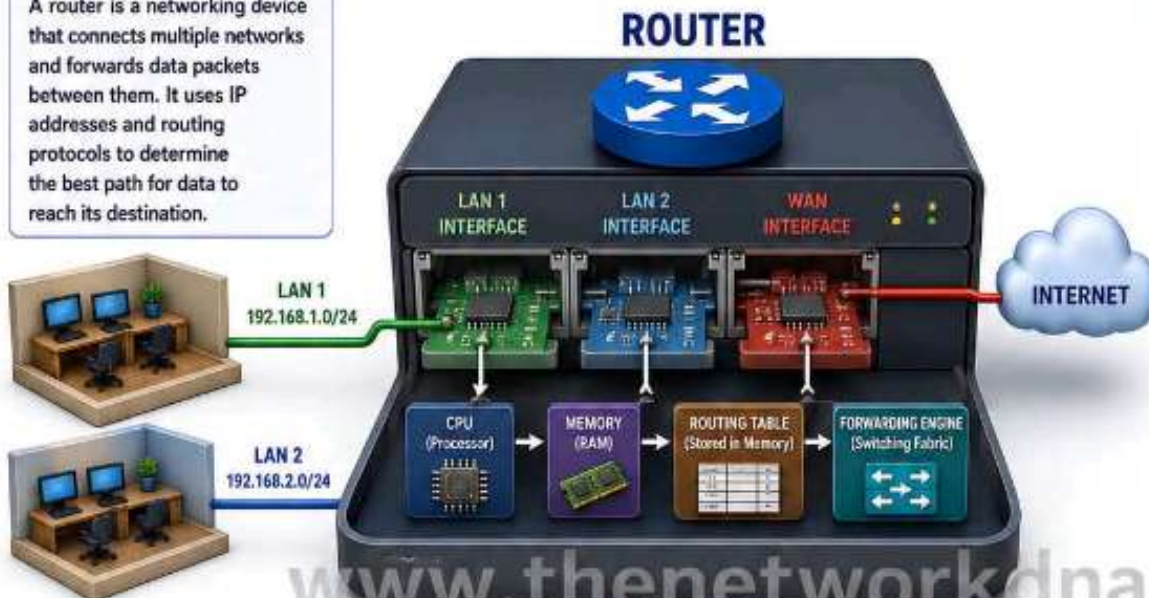
CCNP DataCenter

WAN/Internet



WHAT IS A ROUTER?

A router is a networking device that connects multiple networks and forwards data packets between them. It uses IP addresses and routing protocols to determine the best path for data to reach its destination.



WHAT A ROUTER DOES

- CONNECTS NETWORKS**
Connects multiple networks (LAN, WAN, Internet).
- ROUTES PACKETS**
Examines IP addresses and chooses the best path.
- SEPARATES BROADCAST DOMAINS**
Reduces network traffic by separating broadcast domains.
- SECURITY & CONTROL**
Can enforce access control lists (ACLs) and policies.
- OPTIMIZES PERFORMANCE**
Finds the most efficient path for data delivery.

HOW A ROUTER WORKS (STEP BY STEP)

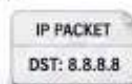
1 RECEIVE PACKET

Router receives a packet on an interface.



2 CHECK DESTINATION IP

Router reads the destination IP address in the packet header.



3 LOOKUP ROUTING TABLE

Router looks up the best path in its routing table.

Destination Network	Next Hop
192.168.1.0/24	LAN 1
192.168.2.0/24	LAN 2
0.0.0.0/0	WAN

4 FORWARD PACKET

Router forwards the packet out the best interface toward the destination.



5 REACH DESTINATION

Packet reaches the destination network or device.



ROUTING TABLE EXAMPLE

Destination Network	Subnet Mask	Next Hop	Interface
192.168.1.0	255.255.255.0	0.0.0.0	LAN 1
192.168.2.0	255.255.255.0	0.0.0.0	LAN 2
10.1.1.0	255.255.255.0	192.168.2.1	LAN 2
0.0.0.0	0.0.0.0	203.0.113.1	WAN

0.0.0.0/0 is the default route (used when no other route matches).

COMMON USES OF ROUTERS

- Connect home/office LAN to the Internet
- Connect multiple branch offices
- Enable communication between different networks
- Support VPN, QoS, and traffic management
- Provide security and access control

TYPES OF ROUTERS



SOHO ROUTER

Used in homes and small offices.



ENTERPRISE ROUTER

Used in large organizations and data centers.



SERVICE PROVIDER ROUTER

Used by ISPs and service providers in backbone networks.



KEY POINTS



Routers work at Layer 3 (Network Layer) of the OSI model.



They use routing protocols like OSPF, EIGRP, BGP, RIP.



Routers make intelligent decisions based on routing tables.



They choose the best path to ensure efficient delivery of data.



They are the backbone of the Internet and modern networks.

OSI MODEL Mindmap

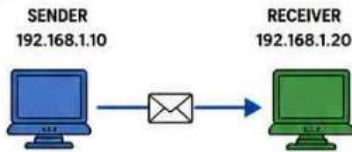


UNICAST, MULTICAST, BROADCAST & ANYCAST

DIFFERENT WAYS DATA IS DELIVERED OVER AN IP NETWORK

1. UNICAST

One sender to one specific receiver.



i Most common type of IP communication.

EXAMPLE:

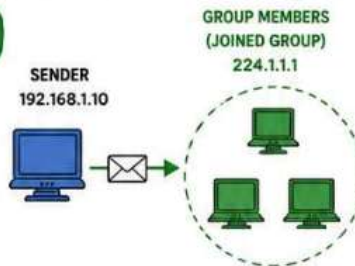
You browsing a website.
Your request goes to the web server, and the server sends response back to you.

USED FOR:

- Web browsing (HTTP/HTTPS)
- Email (SMTP, IMAP)
- File transfer (SSH, FTP)

2. MULTICAST

One sender to a group of specific receivers who have joined the multicast group.



i Saves bandwidth by sending one stream to many receivers.

EXAMPLE:

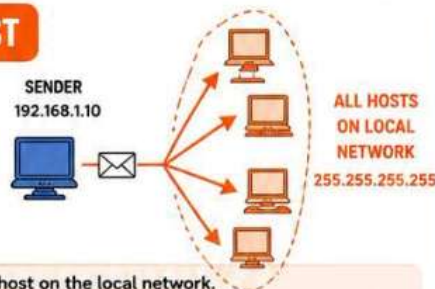
Live video streaming,
online gaming,
stock updates.

USED FOR:

- IPTV
- Video conferencing
- Online gaming
- Real-time data distribution

3. BROADCAST

One sender to all hosts on the local network.



i Reaches every host on the local network.
Can create extra traffic if overused.

EXAMPLE:

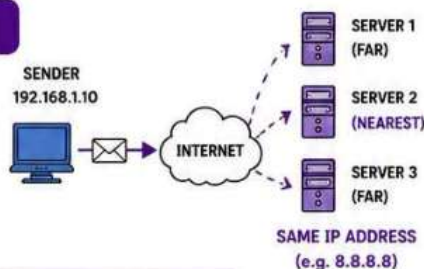
ARP request,
DHCP discovery.

USED FOR:

- Finding devices (ARP)
- Getting IP address (DHCP)
- Routing updates (in some protocols)

4. ANYCAST

One sender to the nearest (best route) receiver from a group of receivers.



i Improves performance and reliability by routing to the nearest available node.

EXAMPLE:

DNS (e.g. Google DNS 8.8.8.8), CDNs.

USED FOR:

- DNS services
- Content Delivery Networks (CDN)
- Load balancing, High availability

TYPE	DELIVERY	RECEIVERS	SCOPE	IP ADDRESS RANGE / EXAMPLE	USE CASE EXAMPLE
UNICAST	One to One	One specific host	Local / Remote	Any regular IP (e.g. 192.168.1.20)	Web browsing, Email
MULTICAST	One to Many (Group)	Group members	Local / Remote	224.0.0.0 – 239.255.255.255	IPTV, Video streaming
BROADCAST	One to All	All hosts on network	Local Network Only	255.255.255.255 (IPv4)	ARP, DHCP discovery
ANYCAST	One to Nearest	Nearest available host	Local / Remote (Internet)	Any regular IP (same IP on many nodes)	DNS, CDN, Load balancing



NetworkDuck
Learn | Practice | Grow

Follow NetworkDuck for more networking tips!



Each method has its own purpose.
Choosing the right one makes networks faster, smarter and more efficient!

#Networking #CCNA #NetworkDuck
#LearnNetworking #IPAddress #Cisco #TechTips

7. STP (SPANNING TREE PROTOCOL)

Protocol : IEEE 802.1D
 BPDU Timer : 2 sec
 Max Age : 20 sec
 Forward Delay : 15 sec

STP Port States:

1. Blocking
2. Listening
3. Learning
4. Forwarding
5. Disabled

8. RSTP (RAPID STP)

Protocol : IEEE 802.1w

States:

1. Discarding
2. Learning
3. Forwarding

9. VLAN (VIRTUAL LAN)

VLAN Range : 1 – 4094
 Normal VLAN : 1 – 1005
 Extended VLAN : 1006 – 4094
 Default VLAN : 1
 Native VLAN : 1

10. ETHERCHANNEL

LACP : 802.3ad
 PagP : Cisco Proprietary

17. ICMP MESSAGE TYPES (COMMON)

TYPE	MEANING
0	Echo Reply
3	Destination Unreachable
5	Redirect
8	Echo Request
11	Time Exceeded

11. HSRP (HOT STANDBY TIME PROTOCOL)

Multicast : 224.0.0.2
 Port : UDP 1985
 Default Priority : 100
 Version 2 Multicast : 224.0.0.102

HSRP States:

1. Initial
2. Listen
3. Learning
4. Speak
5. Standby
5. Active

12. VRRP (VIRTUAL ROUTER REDUNDANCY)

Protocol Number : 112
 Multicast : 224.0.0.18
 Default Priority : 100

13. GLBP (GATEWAY LOAD BALANCING)

Multicast : 224.0.0.102
 Port : UDP 3222
 Default Priority : 100

18. DEFAULT / IMPORTANT TIME VALUES

CDP Timer : 60 sec
 CDP Holdtime : 180 sec
 LLDP Timer : 30 sec
 ARP Timeout : 4 Hours
 MAC Address Table Aging : 300 sec
 CAM Table Aging : 300 sec
 DHCP Lease Default : 1 Day

14. IPv4 IMPORTANT FACTS

Address Size : 32-bit
 Classes : A, B, C, D, E
 Loopback : 127.0.0.0/8
 APIPA : 169.254.0.0/16
 Broadcast : 255.255.255.255

15. IPv6 IMPORTANT FACTS

Address Size : 128-bit
 Loopback : ::1
 Unspecified : :::::
 Link Local : FE80::/10
 Multicast : FF00::/8

16. PRIVATE IP RANGES

NETWORK	CIDR
10.0.0.0	/8
172.16.0.0	/12
192.168.0.0	/16

GOLDEN NUMBERS

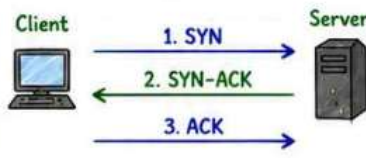
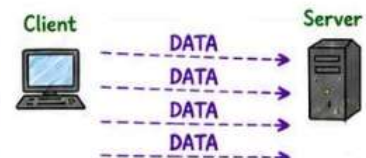
SSH-22, Telnet-23, DNS-53, FTP-69,
 HTTP-80, HTTPS-443, SNMP-161,
 RDP-3389, OSPF AD-110, EIGRP AD-90,
 RRP AD-120, eBGP AD-20, IBGP AD-200,
 VLAN Range - 1-4094, STP Forward Delay -15 sec.
 Max Age - 20 sec, BPDU Timer - 2 sec,
 HSRP Priority -100, VRRP Priority -100,
 MAC Aging - 300 sec.

TCP VS UDP

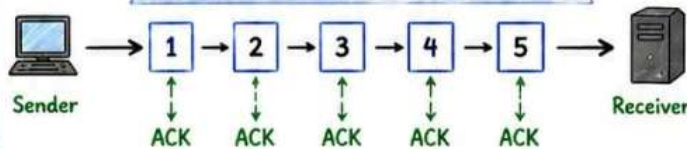
Transmission
Control
Protocol

Same Goal -
Different Approach!

User
Datagram
Protocol

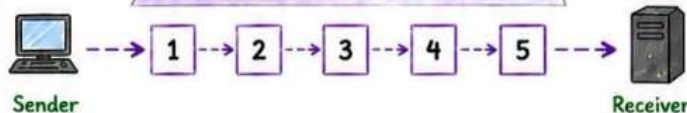
TCP (Reliable)		UDP (Fast)
Connection Oriented (3 - Way Handshake) 	 Connection	Connectionless (No Handshake) 
Reliable - Ensures data reaches in correct order with no loss.	 Reliability	Best Effort - No guarantee of delivery.
Uses Acknowledgement & Retransmission	 Delivery	No Acknowledgement No Retransmission
Flow Control Yes	 Flow Control	No Flow Control
Congestion Control Yes	 Congestion Control	No Congestion Control
Header Size 20 - 60 bytes	 Header Size	Header Size 8 bytes
Speed Slower	 Speed	Speed Faster
Best For Applications where reliability is important. (Web, Email, File Transfer, SSH, FTP, etc.)	 Use Case	Best For Applications where speed is important. (Streaming, Gaming, VoIP, DNS, Video Call, etc.)

TCP - Reliable Data Transfer



Data is divided into segments. All segments are acknowledged. If any segment is lost, it is retransmitted.

UDP - Fast Data Transfer



Data is sent as datagrams. No acknowledgement. No retransmission. Some may be lost, duplicated or arrive out of order.

SUMMARY



TCP is Like a **Tracked Delivery Service**
→ Ensures safe, ordered & reliable delivery. ✓

UDP is Like a **Speed Post**
→ Fast delivery, but no guarantee. 🚀

Choose Wisely Based on Your Application Needs! ❤️



NETWORKING
WITH ISRAR

LEARN | PRACTICE | SUCCEED

INTER-VLAN ROUTING

EXPLAINED

Enabling Communication Between Different VLANs



WHAT IS INTER-VLAN ROUTING?

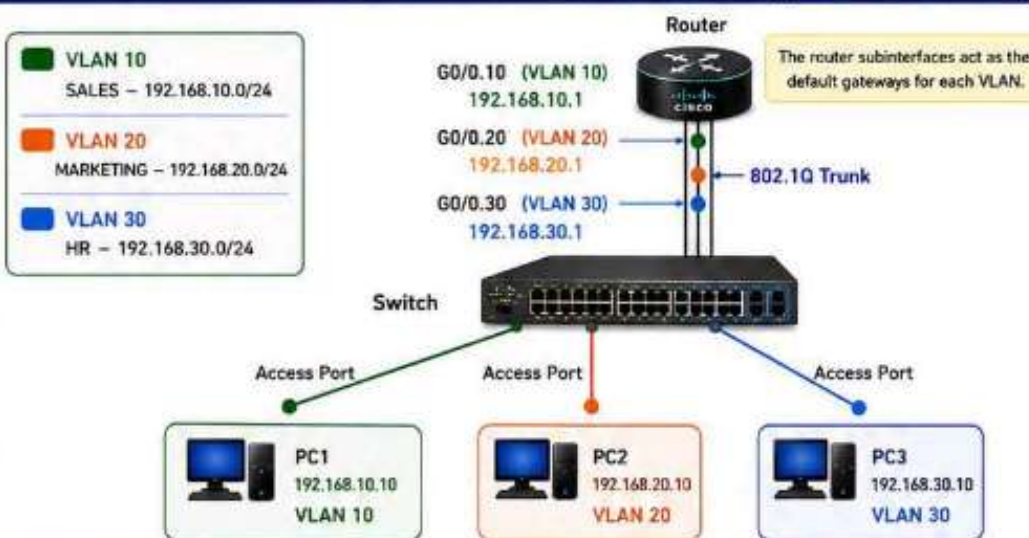
Inter-VLAN routing allows devices in different VLANs to communicate with each other. Since VLANs are separate broadcast domains at Layer 2, a Layer 3 device (Router or Layer 3 Switch) is required to route traffic between them.



WHY IS IT NEEDED?

- ✓ VLANs isolate broadcast domains for security and performance.
- ✓ Devices in different VLANs cannot communicate by default.
- ✓ Inter-VLAN routing enables communication between VLANs.
- ✓ It is essential in enterprise networks.

HOW INTER-VLAN ROUTING WORKS (ROUTER-ON-A-STICK)



HOW COMMUNICATION HAPPENS?

- 1 PC1 (VLAN 10) wants to talk to PC2 (VLAN 20).
- 2 PC1 sends the packet to its default gateway 192.168.10.1 (Router G0/0.10).
- 3 Router receives the packet, removes VLAN 10 tag, checks the destination, and routes it to VLAN 20.
- 4 Router sends the packet out G0/0.20 with VLAN 20 tag.
- 5 Switch receives the tagged packet and forwards it to PC2.

CONFIGURATION EXAMPLE (ROUTER-ON-A-STICK)



Router Configuration

```
interface GigabitEthernet0/0
no ip address
no shutdown
!
interface GigabitEthernet0/0.10
encapsulation dot1Q 10
ip address 192.168.10.1 255.255.255.0
!
interface GigabitEthernet0/0.20
encapsulation dot1Q 20
ip address 192.168.20.1 255.255.255.0
!
interface GigabitEthernet0/0.30
encapsulation dot1Q 30
ip address 192.168.30.1 255.255.255.0
!
```



Switch Configuration (Important Parts)

```
vlan 10
name SALES
!
vlan 20
name MARKETING
!
vlan 30
name HR
!
interface range fa0/1 - 8
switchport mode access
switchport access vlan 10
!
interface range fa0/9 - 16
switchport mode access
switchport access vlan 20
!
interface range fa0/17 - 24
switchport mode access
switchport access vlan 30
!
interface gi0/1
switchport mode trunk
switchport trunk allowed vlan 10,20,30
```

OTHER METHODS



1. LAYER 3 SWITCH (SVI)

SVIs are created on Layer 3 Switch and routing is done internally.



2. DEDICATED ROUTER INTERFACE

Each VLAN connects to a separate router interface (legacy method).

KEY POINTS

- ✓ VLANs = Separate Broadcast Domains
- ✓ Inter-VLAN Routing = Communication Between VLANs
- ✓ Layer 3 Device is Required
- ✓ Router-on-a-Stick is a Common Method
- ✓ 802.1Q Trunk Carries Multiple VLANs



NETWORKING WITH ISRAR

LEARN | PRACTICE | SUCCEED

• Garansi nya full semua • Bukan akun trial ini akun privat • Bisa pake email kalian



Claude Pro
125K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



Claude Max
175K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



ChatGPT Plus
75K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



ChatGPT Pro
100K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



ChatGPT Business Codex
135K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



Gemini Pro
120K

- 18 Bulan
- Full garansi
- Privat
- Bisa email sendiri



Kiro Pro
60K

- 1 Bulan
- Full garansi
- Credit 1000
- Privat
- Bisa email sendiri atau GitHub



Kiro Power
100K

- 1 Bulan
- Credit 10.000
- Login GitHub (bawa sendiri bisa)



DeepSeek
130K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



Leonardo AI
60K

- 3 Bulan
- Full garansi
- Privat
- Bisa email sendiri



DOLA Unlimited
99K

- Selamanya
- Full garansi
- Privat
- Bisa email sendiri



SuperGrok
175K

- 3 Bulan
- Full garansi
- Privat
- Bisa email sendiri



MIMO Xiaomi
100K

- 100M Credit
- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



Google One Ultra
135K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri



Cursor
105K

- 1 Bulan
- Full garansi
- Privat
- Bisa email sendiri

TINGKATKAN PRODUKTIVITAS MU DENGAN AI TERBAIK!



CHAT ADMIN SEKARANG!



PROSES CEPAT LANGSUNG AKTIF



AMAN 100% PRIVAT & TERPERCAYA

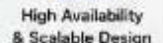


SUPPORT 24/7 SIAP BANTU KAPAN SAJA



HARGA TERBAIK KUALITAS PREMIUM

VLANs + INTER-VLAN ROUTING + EIGRP + NAT + DHCP



SEGMENT	NETWORK	GATEWAY / INTERFACE	DESCRIPTION
R1 - R2 Link	10.0.0.0/30	R1: 10.0.0.1 R2: 10.0.0.2	WAN Link
R2 - R3 Link	10.0.0.4/30	R2: 10.0.0.5 R3: 10.0.0.6	WAN Link
R2 - Core LAN	192.168.1.0/24	R2 G0/2: 192.168.1.1	Core Management LAN
Branch LAN (R3)	192.168.3.0/24	R3 G0/0: 192.168.3.1	Branch Office LAN
VLAN 10 (SALES)	192.168.10.0/24	SW1 G0/0.10: 192.168.10.1	Users
VLAN 20 (FINANCE)	192.168.20.0/24	SW1 G0/0.20: 192.168.20.1	Users
VLAN 30 (ENGINEERING)	192.168.30.0/24	SW1 G0/0.30: 192.168.30.1	Users
VLAN 40 (HR)	192.168.40.0/24	SW1 G0/0.40: 192.168.40.1	Users
VLAN 50 (MANAGEMENT)	192.168.50.0/24	SW1 G0/0.50: 192.168.50.1	Users
VLAN 99 (SERVERS)	192.168.99.0/24	SW1 G0/0.99: 192.168.99.1	Servers



NETWORK SECURITY – CHEAT SHEET / ROADMAP

NETWORK FUNDAMENTALS

- OSI Model
- TCP/IP Model
- IP Addressing (IPv4/IPv6)
- Subnetting / CIDR
- VLANs, Trunks, STP
- Routing Basics (Static, Default)
- Protocols: ARP, DNS, ICMP, OSPF
- Common Ports (TCP/UDP, SSH, Telnet, RDP, LDAP, etc.)

ROUTING & SWITCHING

- Dynamic Routing: OSPF, EIGRP, BGP
- Policy Based Routing (PBR)
- Route Redistribution
- VRF, MPLS Basics (L3VPN)
- HSRP/VRRP/GLBP
- EtherChannel (LACP)
- VLAN, VTP, PVST+, MSTP

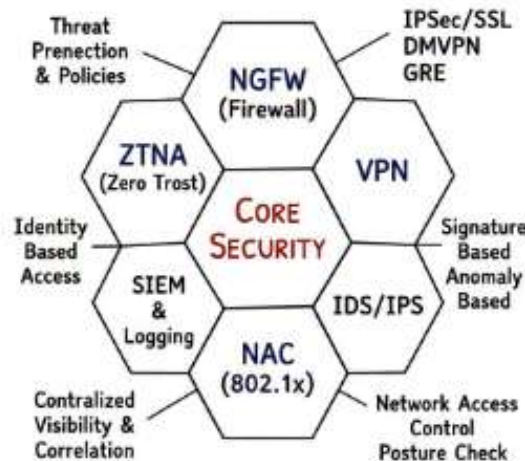
NETWORK SECURITY BASICS

- CIA Triad: Confidentiality, Integrity, Availability
- AAA: Authentication, Authorization, Accounting
- Access Control: ACLs
- Security Policies & Baselines
- Threats, Vulnerabilities, Exploits
- Defense in Depth

PERIMETER SECURITY

- Next-Generation Firewalls (NGFW)
- NAT, PAT, Static NAT
- DMZ Design
- VPN (Site-to-Site, Remote Access)
- Proxy, Content Filtering
- IDS / IPS

CORE SECURITY TECHNOLOGIES



SECURITY PROTOCOLS & SERVICES

- AAA: RADIUS, TACACS+
- 802.1X (Port-Based Access Control)
- NTP, SNMPv3, Spring
- SSH, HTTPS, SCP, SFTP
- IPsec (IKEv1 / IKEv2)
- TLS / SSL, DTLS
- PKI, Certificates, Digital Signatures

ADVANCED SECURITY

- Zero Trust Network Access (ZTNA)
- SASE (Secure Access Service Edge)
- NAC - Network Access Control
- Micro-segmentation
- DLP (Data Loss Prevention)
- SWG (Secure Web Gateway)
- CASB (Cloud Access Security Broker)
- XDR / EDR / NDR
- Threat Intelligence

FIREWALL TECHNOLOGIES

- Palo Alto Networks (PAN-OS)
- Fortinet FortiGate (FortiOS)
- Cisco Firepower (FTO)
- Cisco ASA (Legacy)
- Check Point
- Firewall Policies (Allow, Deny)
- NAT Policies, Security Zones
- Application Control, IPS, AV
- SSL Inspection, Decryption

AAA & ACCESS CONTROL

- TACACS+, RADIUS
- Role Based Access Control (RBAC)
- 802.1X / EAP-TLS
- Certificates (PKI)
- Cisco ISE / Aruba ClearPass

CLOUD NETWORKING & SECURITY

- AWS: VPC, Subnets, SG, NACL, TGM, Direct Connect, VPN, Route 53
- Azure: VNet, Azure Monitor
- GCP: VPC, Subnets, Firewall Rules, Cloud VPN, Cloud Router, Cloud NAT, Cloud DNS, Load Balancing
- Cloud Security Best Practices
- Hybrid Connectivity, Interconnect

MONITORING & LOGGING

- SIEM Platforms: Splunk, Microsoft Sentinel, QRadar
- Network Monitoring: SolarWinds, PRTG
- Log Analysis, Alerting
- NetFlow, eFlow
- Packet Capture: Wireshark, tcpdump

COMPLIANCE & STANDARDS

- ISO 27001
- PCI-DSS
- NIST 800-53
- GDPR

ROADMAP (STEP-BY-STEP)

Build Strong Networking Fundamentals
Learn Routing & Switching in depth
Understand Network Security Basics
Learn Firewalls, VPNs, and Perimeter Security
Master AAA, ACLs, and Security Protocols
Hands-on Advanced Security Technologies
Hands-on with Security Tools & Monitoring
Learn Cloud & Modern Security Concepts
Work on Real-world Scenarios / Labs
Get Certified & Keep Learning

LAB IDEAS

- Build a small enterprise network in GNS3 / EVE-NG / Packet Tracer
- Configure Firewall, NAT, VPN
- Implement ACLs and 802.1X
- Set up SIEM Lab with Splunk (Free)
- Simulate Attacks & analyze logs

IMPORTANT CONCEPTS

- Least Privilege Access
- Always Verify, Never Trust
- Encrypt Everything
- Monitor Continuously
- Patch Regularly
- Backup & Test Recovery

USEFUL RESOURCES

- Cisco Networking Academy
- Palo Alto EBU-210
- Fortinet NSE Training
- TryHackMe / HackTheBox
- Wireshark University
- NIST Cybersecurity Framework
- SANS (Free Resource)
- YouTube: NetworkChuck, CBT Nuggets

KEY FORMULA

Subnet Mask = $2^n - 2$
where n = (32 - Prefix Length)

FILTER INBOUND. FILTER OUTBOUND. EVERY UNFILTERED SESSION IS A RISK.

BGP ROUTE FILTERING MASTERCLASS

1 PREFIX-LIST

Control what you accept or advertise

✗ WITHOUT FILTER

900,000 ROUTES RECEIVED FROM ISP
(ROUTER MEMORY EXHAUSTED)



✓ WITH ACCEPT-DEFAULT-ONLY PREFIX-LIST

1 ROUTE RECEIVED
(DEFAULT ROUTE ONLY)



PREFIX-LIST EXAMPLE

```
ip prefix-list ACCEPT-DEFAULT-ONLY seq 5 permit 0.0.0.0/0
!
router bgp 65001
neighbor 203.0.113.1 prefix-list ACCEPT-DEFAULT-ONLY in
```

ge / le EXPLAINED

Control prefix length range



PERMIT 10.0.0.0/8 ge 16 le 24

PREFIX	LENGTH	MATCH?
10.0.0.0/8	/8	✗ Too General
10.1.0.0/16	/16	✓ ACCEPT
10.1.1.0/24	/24	✓ ACCEPT
10.1.1.128/25	/25	✗ Too Specific

ge = greater or equal (more specific)

le = less or equal (less specific)

2 AS_PATH ACL

Filter based on AS path attributes

^ \$ (Empty AS_PATH)

Matches empty AS_PATH =
your own originated routes only



Use case: Advertise only your own
routes to the Internet

^65002\$ (Exact Match)

Matches routes where AS_PATH
contains ONLY 65002 (direct routes)



Use case: Accept only routes directly
originated by your peer

65002 (Underscore)

Matches routes that TRANSIT through
AS 65002 (any position in path)



Use case: Filter routes going through
a specific transit AS

COMMON REGEX ELEMENTS

- ^ Start of AS_PATH
- \$ End of AS_PATH
- _ Match anywhere in AS_PATH
- | OR operator
- () Grouping

BGP uses a regular expression
engine for AS_PATH matching

3 ROUTE-MAP

Match + Set = Powerful Control

Use route-maps to match conditions
and set attributes in one rule.

EXAMPLE: Set LOCAL_PREF for inbound routes

```
ip prefix-list CUSTOMER-NETS seq 5 permit 10.0.0.0/8 le 24
ip prefix-list CUSTOMER-NETS seq 10 permit 172.16.0.0/12 le 24
!
route-map INBOUND-FROM-ISP permit 10
match ip address prefix-list CUSTOMER-NETS
set local-preference 200
!
router bgp 65001
neighbor 203.0.113.1 route-map INBOUND-FROM-ISP in
```

HOW IT WORKS



MATCH

prefix-list
CUSTOMER-NETS

SET

local-preference 200

RESULT

Routes from ISP with our prefixes
get higher priority

ROUTE-MAP CAPABILITIES

- ✓ Set LOCAL_PREF (outbound traffic control)
- ✓ Set MED (inbound traffic control)
- ✓ Set communities (advanced policies)
- ✓ Filter routes (permit / deny)
- ✓ Modify next-hop, weight, tag, etc.

ROUTE LEAK SCENARIO (DANGER!)

Misconfiguration can disrupt the entire Internet

THE PROBLEM

You accidentally advertise your full routing table to the ISP.
This can cause routing loops, blackholes, or hijacks.

YOUR NETWORK

AS 65001
10.0.0.0/8
172.16.0.0/12
192.168.0.0/16
+ 800,000 more routes

ACCIDENTALLY
ADVERTISED



BAD CONFIG EXAMPLE

```
router bgp 65001
neighbor 203.0.113.1 route-as 64500
neighbor 203.0.113.1 advertise 12vpn evpn
neighbor 203.0.113.1 network 0.0.0.0 max-prefix 0
! (Advertising everything!)
```



NEVER
DO THIS!

INTERNET IMPACT

- Routing table contamination
- Traffic disruption
- Possible BGP route hijack
- Reputation damage

ALWAYS FILTER WHAT YOU ADVERTISE!

5 BOGON FILTER LIST (BLOCK THESE!)

Never accept or advertise these prefixes

TYPE	PREFIX	DESCRIPTION	ACTION
RFC 1918 (Private)	10.0.0.0/8 172.16.0.0/12 192.168.0.0/16	Private IP ranges (Not for Internet)	✗ BLOCK
Loopback	127.0.0.0/8	Loopback range	✗ BLOCK
Link-Local	169.254.0.0/16	Link-local range	✗ BLOCK
Carrier-Grade NAT	100.64.0.0/10	CGNAT range (RFC 6598)	✗ BLOCK
Reserved	0.0.0.0/8 255.255.255.255/32 224.0.0.0/4 240.0.0.0/4	Reserved / Multicast / Special use	✗ BLOCK
Documentation	192.0.2.0/24 198.51.100.0/24 203.0.113.0/24	Documentation ranges (RFC 5737)	✗ BLOCK

Use inbound prefix-lists or route-maps to block bogons!

KEY TAKEAWAYS

- ✓ Filter inbound routes to protect your router and network.
- ✓ Filter outbound routes to control how you are seen.
- ✓ Use PREFIX-LIST, AS_PATH ACL, and ROUTE-MAP effectively.
- ✓ Always test your filters before applying in production!

GOLDEN RULE

**CONTROL WHAT COMES IN.
CONTROL WHAT GOES OUT.
CONTROL YOUR NETWORK.**



Understanding Computer Ports

What is USB?

USB stands for *peripherals* and transferring data.



Ethernet

For network connections and internet access.



HDMI

For connecting monitors, TVs, and projectors.



DisplayPort

Also for monitors.



Audio Jacks

For headphones, microphones, and speakers.



DCE COMPUTER

Top 50 Networking Protocol Interview

Questions with Answers (Part 2)

11. What is HTTPS?

Ans: Secure HTTP using SSL/TLS encryption.

Port: 443

Example: <https://google.com>

12. Difference between HTTP and HTTPS?

HTTP	HTTPS
Not encrypted	Encrypted
Port 80	Port 443
Less secure	More secure

13. What is FTP?

Ans: File Transfer Protocol transfers files.

Port: 21

Example: Uploading website files.

14. What is SFTP?

Ans: Secure File Transfer over SSH.

Port: 22

Example: Secure file upload.

15. What is TFTP?

Ans: Trivial File Transfer Protocol. Uses UDP.

Port: 69

Example: Router firmware update.

16. What is SMTP?

Ans: Simple Mail Transfer Protocol sends emails.

Port: 25

Example: Sending Gmail.

17. What is POP3?

Ans: Downloads email from server.

Port: 110

Example: Outlook downloading mail.

18. What is IMAP?

Ans: Keeps emails on the server.

Port: 143

Example: Accessing same inbox on phone and laptop.

19. Difference between POP3 and IMAP?

POP3	IMAP
Downloads emails	Synchronizes emails
Local storage	Server storage

20. What is DNS?

Ans: Domain Name System converts names to IP addresses.

Example: google.com

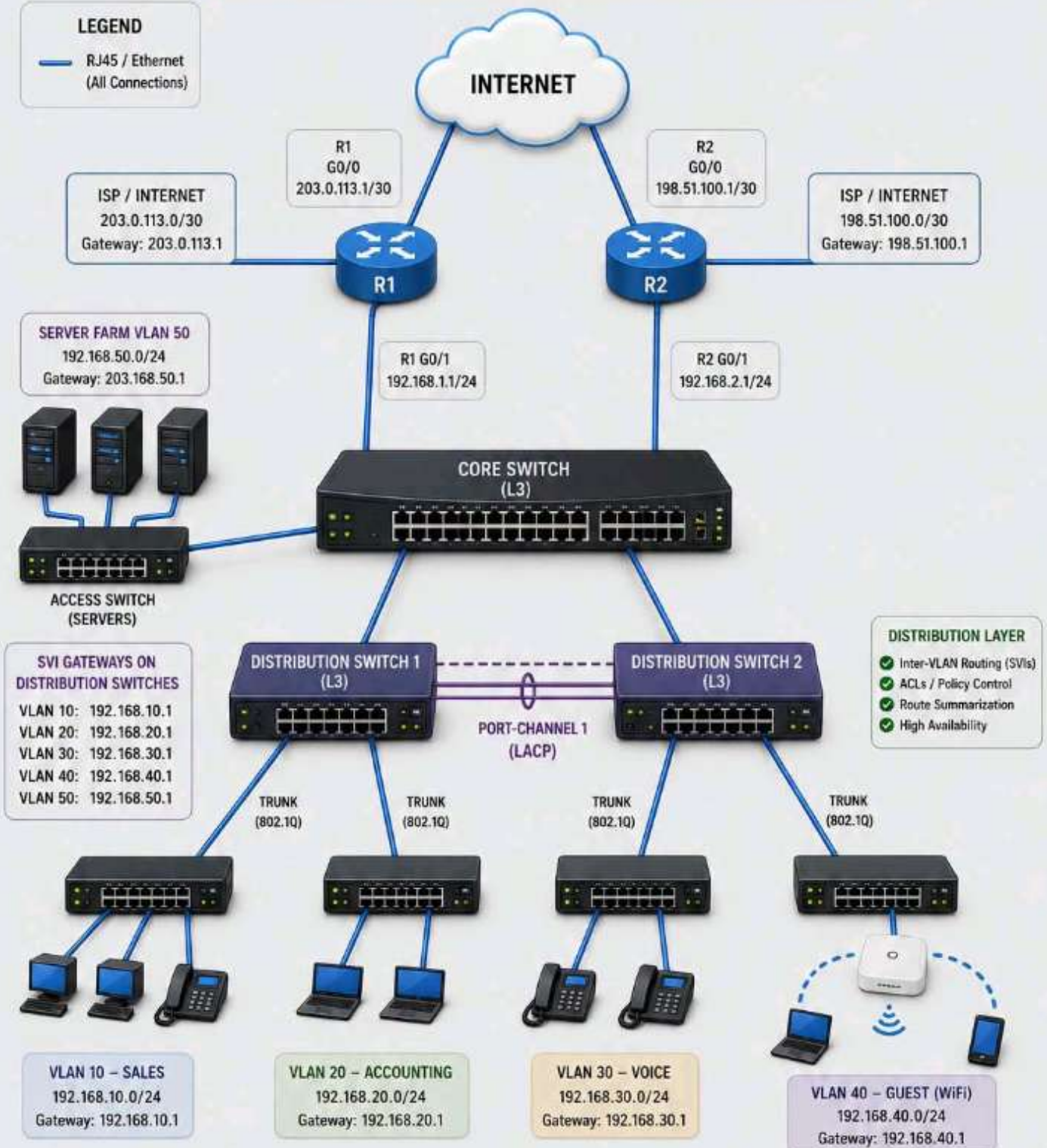
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142.250.xxx.xxx

CCNA NETWORK TOPOLOGY

LEARN • CONFIGURE • TROUBLESHOOT • SUCCEED

LEGEND

— RJ45 / Ethernet
(All Connections)



HIGH AVAILABILITY

- Redundant Links
- LACP Port-Channel
- Fast Convergence
- Dual Distribution



PERFORMANCE

- Optimized Routing
- Efficient Traffic Flow
- Load Balancing
- QoS Ready



SECURITY

- VLAN Segmentation
- ACLs
- Port Security
- Secure Management



SCALABILITY

- Modular Design
- Easy Expansion
- Future Ready
- Supports Growth

ADDRESSING SUMMARY

VLAN 10 – SALES	192.168.10.0/24	GW: 192.168.10.1	VLAN 40 – GUEST (WiFi)	192.168.40.0/24	GW: 192.168.40.1
VLAN 20 – ACCOUNTING	192.168.20.0/24	GW: 192.168.20.1	VLAN 50 – SERVER FARM	192.168.50.0/24	GW: 192.168.50.1
VLAN 30 – VOICE	192.168.30.0/24	GW: 192.168.30.1			



www.thenetworkdna.com

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Reserved	0.0.0.0/8 255.255.255.255/32 224.0.0.0/4 240.0.0.0/4	Reserved / Multicast / Special use	✗ BLOCK
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Claude for Project Managers in 28 days

Day 1

Claude Basics for PMs
(22 min)

Day 2

Project Charter & Scope Definition
(25 min)

Day 3-4

Risk Management & Resource Planning
(21 min)

Day 5

Stakeholder Updates & Reporting
(27 min)

Day 6

Meeting Prep & Action Tracking
(26 min)

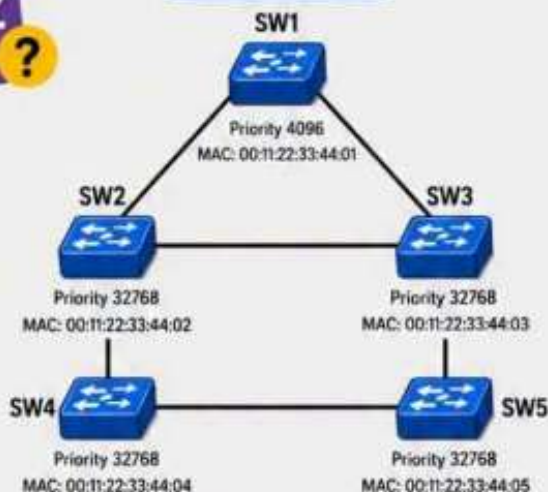
Day 7

Advanced Workflow Automation
(41 min)

Certificate on completion. 28 days, 15 min/day. Self-paced.

QUIZ TIME ?

Network Diagram



SPANNING TREE PROTOCOL (STP)

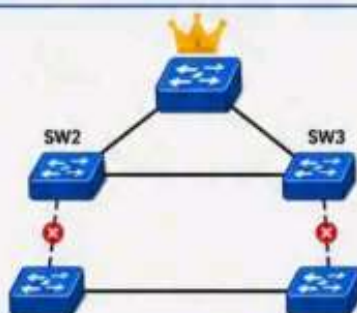
Based on default STP operation, which switch will be the Root Bridge and which links will be in blocking state?

A

Root Bridge: SW1

Blocking Links:

- SW2 – SW4 link
- SW3 – SW5 link



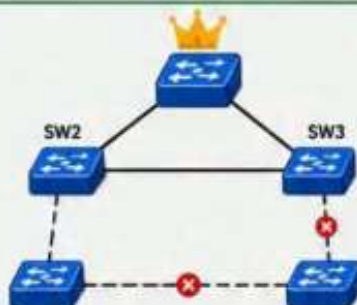
- ✓ SW1 has the lowest Bridge ID (lowest priority).
- ✓ SW2 and SW3 have equal cost to root, but SW2 has lower BID than SW3, so it wins the tie to SW4.
- ✓ SW3 – SW5 link becomes blocking.

B

Root Bridge: SW1

Blocking Links:

- SW4 – SW5 link
- SW3 – SW5 link



- ✓ SW1 has the lowest Bridge ID (root bridge).
- ✓ All links from root to SW2 and SW3 are forwarding.
- ✓ SW2 is the designated bridge for SW4.
- ✓ SW4 – SW5 link becomes blocking.

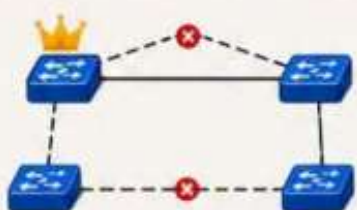
BEST CHOICE

C

Root Bridge: SW2

Blocking Links:

- SW1 – SW3 link
- SW4 – SW5 link



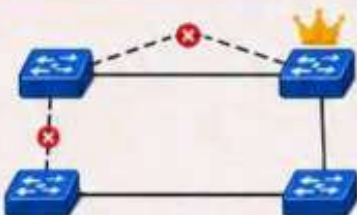
- ✓ SW2 has lower Bridge ID than SW1.
- ✓ Top link between SW1 and SW3 becomes blocking.
- ✓ SW4 – SW5 link becomes blocking.

D

Root Bridge: SW3

Blocking Links:

- SW1 – SW2 link
- SW2 – SW4 link



- ✓ SW3 has the lowest Bridge ID.
- ✓ Links from SW1 and SW2 will be blocked.
- ✓ SW2 – SW4 link becomes blocking.

STP KEY FACTS



Root Bridge

The switch with the lowest Bridge ID (becomes root).



Root Ports

Each non-root switch has one root port with the lowest path cost to root.



Designated Ports

One designated port per segment forwards traffic.



Blocking Ports

Extra ports are blocked to prevent loops.

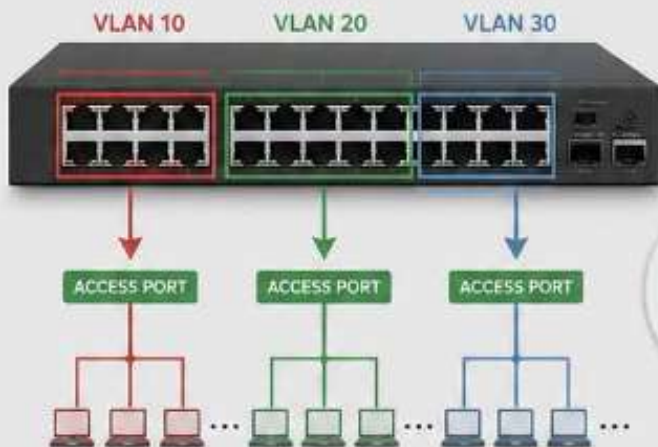


STP Benefits

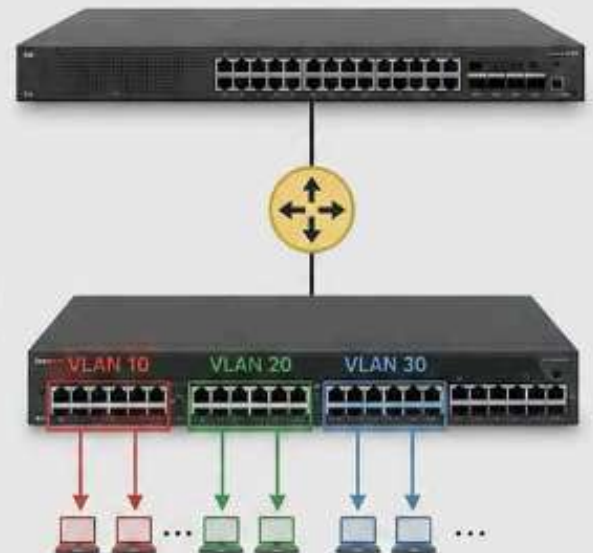
Prevents loops, ensures redundancy, and provides high availability.

Protocol	Port Number(s)	Transport Protocol	Description
HTTP	80	TCP	Web traffic
HTTPS	443	TCP	Secure web traffic
FTP (Data)	20	TCP	File Transfer (data)
FTP (Control)	21	TCP	File Transfer (control)
SFTP	22	TCP	Secure File Transfer (over SSH)
SSH	22	TCP	Secure remote login
Telnet	23	TCP	Remote login (unencrypted)
SMTP	25	TCP	Email sending
POP3	110	TCP	Email retrieval
IMAP	143	TCP	Email retrieval (advanced)
LDAP	389	TCP / UDP	Directory services
LDAPS	636	TCP	Secure LDAP
DNS	53	TCP / UDP	Domain Name Resolution
DHCP (Server)	67	UDP	Assigns IP addresses
DHCP (Client)	68	UDP	Receives IP addresses
TFTP	69	UDP	Trivial file transfer
SNMP (Request)	161	UDP	Network management
SQL Server(MSSQL)	1433	TCP	Database service
Oracle Database	1521	TCP	Database service

LAYER 2 SWITCH



LAYER 3 SWITCH



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- ✓ Operates on **Data Link** layer of the OSI model.
- ✓ Sends frames to destination based on the **MAC** addresses.
- ✓ Works with **MAC** addresses only.
- ✓ Used to reduce traffic on the **Local Network**.
- ✓ Operate within a **Single Broadcast** domain.
- ✓ **Faster** than Layer 3 switches as they don't examine the data packets.

- ✓ Operates on the **Network** Layer of the OSI model.
- ✓ **Routes** packets to the destination based on the **IP** addresses.
- ✓ Can perform function of Layer 2 & Layer 3 switches.
- ✓ Has multiple **Broadcast** domains.
- ✓ **Slower** than Layer 2 switches as to examine data packets.

20 WEBSITES **Google** DOESN'T WANT YOU TO KNOW ABOUT



1. <http://12ft.io> — Bypass any paywall
2. <http://libgen.is> — Millions of free textbooks
3. <http://sci-hub.se> — Free research papers
4. <http://alternativeto.net> — Find free app alternatives
5. <http://justwatch.com> — Find streaming locations for any content
6. <http://archive.org> — Access any old webpage
7. <http://gutenberg.org> — 70,000 free classic books
8. <http://pdfdrive.com> — Free PDF downloads
9. <http://openculture.com> — Free courses from top universities
10. <http://wolframalpha.com> — Instantly solve any math problem
11. <http://photopea.com> — Free Photoshop in your browser
12. <http://squoosh.app> — Compress any image for free
13. <http://remove.bg> — Remove image backgrounds for free
14. <http://cleanup.pictures> — Erase objects from photos
15. <http://unscreen.com> — Remove any video backgrounds
16. <http://carbon.now.sh> — Turn code into art
17. <http://ray.so> — Beautiful code screenshots
18. <http://shots.so> — Free product mockups
19. <http://smartmockups.com> — Mockups without Photoshop
20. <http://haveibeenpwned.com> — Check if you've been hacked

BGP ADVANCED CONFIGURATION LAB

CONFIGURE BGP WITH LOOPBACK, NEXT-HOP AND EBGP MULTI-HOP

OBJECTIVE

Configure advanced BGP using loopback as update source, enable next-hop self and allow eBGP multi-hop between neighbors.

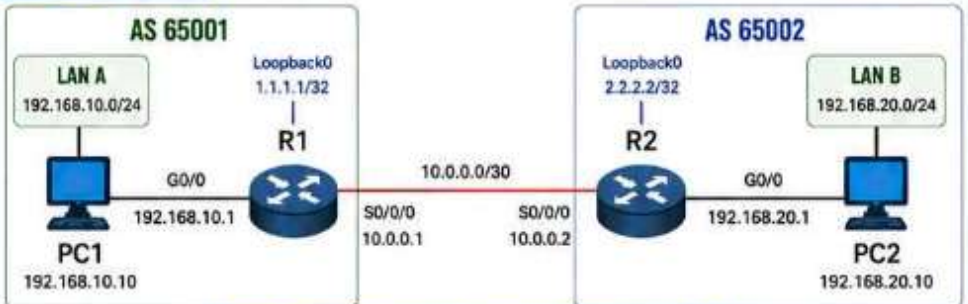


TASKS

1. Configure loopback interfaces.
2. Establish eBGP using loopbacks.
3. Enable eBGP multi-hop.
4. Enable next-hop-self on R2.
5. Advertise LAN networks.
6. Verify routes and next-hop.



TOPOLOGY



eBGP Neighbor: R1(1.1.1.1) ↔ R2(2.2.2.2) using loopbacks

IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Description
R1	G0/0	192.168.10.1	255.255.255.0	LAN A Gateway
R1	S0/0/0	10.0.0.1	255.255.255.252	Link to R2
R1	Loopback0	1.1.1.1	255.255.255.255	BGP Update Source
R2	S0/0/0	10.0.0.2	255.255.255.252	Link to R1
R2	G0/0	192.168.20.1	255.255.255.0	LAN B Gateway
R2	Loopback0	2.2.2.2	255.255.255.255	BGP Update Source

R1 - CONFIGURATION (AS 65001)

```
1 enable
2 configure terminal
3 hostname R1
4 interface g0/0
5 ip address 192.168.10.1 255.255.255.0
6 no shutdown
7 interface s0/0/0
8 ip address 10.0.0.1 255.255.255.252
9 no shutdown
10 interface loopback0
11 ip address 1.1.1.1 255.255.255.255
12 router bgp 65001
13 neighbor 2.2.2.2 remote-as 65002
14 neighbor 2.2.2.2 update-source loopback0
15 neighbor 2.2.2.2 ebgp-multi-hop 5
16 network 192.168.10.0 mask 255.255.255.0
17 end
```

R2 - CONFIGURATION (AS 65002)

```
1 enable
2 configure terminal
3 hostname R2
4 interface g0/0
5 ip address 192.168.20.1 255.255.255.0
6 no shutdown
7 interface s0/0/0
8 ip address 10.0.0.2 255.255.255.252
9 no shutdown
10 interface loopback0
11 ip address 2.2.2.2 255.255.255.255
12 router bgp 65002
13 neighbor 1.1.1.1 remote-as 65001
14 neighbor 1.1.1.1 update-source loopback0
15 neighbor 1.1.1.1 ebgp-multi-hop 5
16 neighbor 1.1.1.1 next-hop-self
17 network 192.168.20.0 mask 255.255.255.0
18 end
```

VERIFICATION COMMANDS

- show ip bgp summary
- show ip bgp
- show ip route bgp
- show ip protocols
- show ip cef | include 192.168
- ping 192.168.20.10 source 192.168.10.10 (from PC1)



These commands help you verify BGP sessions, routes and next-hop reachability.

HOW IT WORKS



1. BGP session is established using loopback addresses.
2. eBGP multi-hop allows the session over multiple hops.
3. Next-hop-self on R2 ensures R1 uses R2 as next-hop.
4. LAN A and LAN B routes are exchanged via BGP.

★ Loops provide stability – BGP stays up even if physical interfaces change.

LEARNING POINTS

- Loopback interfaces are ideal for BGP update sources.
- eBGP multi-hop is required when neighbors are not directly connected.
- Next-hop-self ensures packets use the neighbor as the next hop.
- Keep BGP sessions stable and independent of physical links.
- Always verify routes, next-hop and reachability.

CHALLENGE QUESTION



If R2 does not configure "next-hop-self", what will be the next-hop IP for the routes received by R1?

How will R1 reach 192.168.20.10 in that case? Will it be able to reach? Explain.



Comment your answer below!

QUICK TEST



Can PC1 ping PC2 successfully?



Is BGP session established?



Are LAN A and LAN B visible in BGP table?



Is next-hop for LAN B routes showing R2 (2.2.2.2) on R1?



Does connectivity remain if serial link flaps but loopbacks stay up?

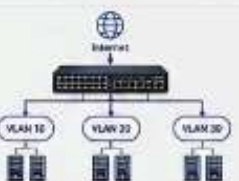




TYPES OF NETWORK SWITCHES



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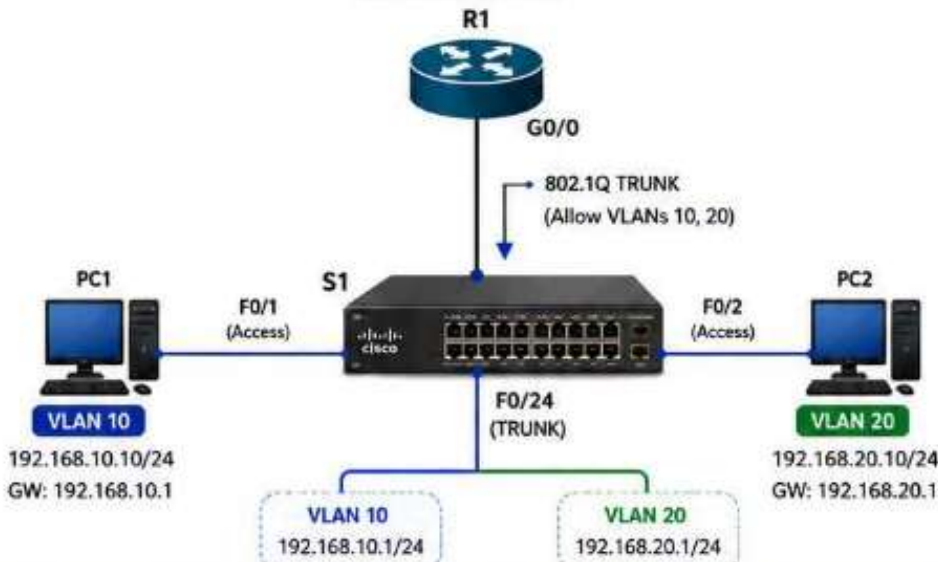
SWITCH TYPE	TOP VIEW	KEY FEATURES	BEST FOR	USE CASE DIAGRAM		
1 UNMANAGED SWITCH  Plug-and-play with no configuration. Ideal for basic connectivity.		✓ No configuration needed ✓ Plug-and-play ✓ Cost-effective ✓ 8/16/24/48 ports options	 Home networks Small offices Basic connectivity			
2 MANAGED SWITCH  Advanced features with configuration and management options.		✓ VLAN, QoS, LACP, STP ✓ SNMP monitoring ✓ Port mirroring ✓ Better security & control ✓ Web/CLI management	 SMBs Growing networks Mission-critical applications			
3 POE SWITCH (Power over Ethernet)  Provides power and data over Ethernet cables to power devices.		✓ Power + Data over Ethernet ✓ Supports PoE/PoE+/PoE++ ✓ Ideal for IP devices ✓ Uplink SFP options	 IP Cameras VoIP Phones Wireless APs IoT Devices			
4 LAYER 2 SWITCH  Operates at OSI Layer 2. High performance and advanced traffic control.		✓ VLAN, STP, QoS ✓ Link Aggregation ✓ IGMP Snooping ✓ MAC Address Table Management ✓ High performance	 Enterprise networks Departmental networks Campus networks			
5 LAYER 3 SWITCH  Advanced switching with routing capabilities between VLANs.		✓ Layer 2 + Layer 3 features ✓ Inter-VLAN Routing ✓ Static & Dynamic Routing ✓ ACL, QoS, Security ✓ High scalability	 Large enterprises Data centers Complex networks			
6 MODULAR SWITCH  Highly scalable with modular components and flexible interfaces.		✓ Modular design ✓ Hot-swappable modules ✓ High port density ✓ Redundant power & fans ✓ Built for high availability	 Data centers Service providers High-density environments			
QUICK COMPARISON						
TYPE	UNMANAGED SWITCH	MANAGED SWITCH	POE SWITCH	LAYER 2 SWITCH	LAYER 3 SWITCH	MODULAR SWITCH
MANAGEMENT	No	Yes	Yes	Yes	Yes	Yes
LAYER	Layer 2	Layer 2	Layer 2	Layer 2	Layer 2 + 3	Layer 2 + 3
POE SUPPORT	No	Optional	Yes	Optional	Optional	Optional
ROUTING	No	No	No	No	Yes	Yes
BEST FOR	Home / Small Office	SMB / Enterprise	IP Devices / IoT	Enterprise Networks	Large Enterprises	Data Centers / SPs
					 PRO TIP Choose the right switch based on your current needs and future scalability. Managed and Layer 3 switches provide better control, security, and performance for growing networks.	

ROUTER ON A STICK LAB



INTER-VLAN ROUTING USING A SINGLE PHYSICAL INTERFACE

LAB TOPOLOGY



LAB OBJECTIVE

Enable inter-VLAN routing using a single physical interface on the router through subinterfaces.

VLAN SUMMARY

VLAN	NAME	SUBNET	GATEWAY
10	SALES	192.168.10.0/24	192.168.20.1
20	MARKETING	192.168.20.0/24	192.168.20.1

PC1 CONFIGURATION (VLAN 10)



- IP Address : 192.168.10.10
- Subnet Mask : 255.255.255.0
- Default Gateway : 192.168.10.1

PC2 CONFIGURATION (VLAN 20)



- IP Address : 192.168.20.10
- Subnet Mask : 255.255.255.0
- Default Gateway : 192.168.20.1

1 CREATE VLANs

```
Switch(config)# vlan 10
Switch(config-vlan)# name SALES
!
Switch(config)# vlan 20
Switch(config-vlan)# name MARKETING
!
Switch(config)# end
Switch# write memory
```

2 ASSIGN ACCESS PORTS

PC1 PORT (F0/1)

```
Switch(config)# interface f0/1
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 10
Switch(config-if)# no shutdown
```

PC2 PORT (F0/2)

```
Switch(config)# interface f0/2
Switch(config-if)# switchport mode access
Switch(config-if)# switchport access vlan 20
Switch(config-if)# no shutdown
```

3 CONFIGURE TRUNK PORT

```
Switch(config)# interface f0/24
Switch(config-if)# switchport mode trunk
Switch(config-if)# switchport trunk
    allowed vlan 10,20
Switch(config-if)# no shutdown

Switch(config-if)# end
Switch# write memory
```

ROUTER ON A STICK CONFIGURATION (SUBINTERFACES)

VLAN 10 - SUBINTERFACE

```
Router(config)# interface g0/0.10
Router(config-subif)# encapsulation dot1q 10
Router(config-subif)# ip address 192.168.10.1 255.255.255.0
Router(config-subif)# no shutdown
```

VLAN 20 - SUBINTERFACE

```
Router(config)# interface g0/0.20
Router(config-subif)# encapsulation dot1q 20
Router(config-subif)# ip address 192.168.20.1 255.255.255.0
Router(config-subif)# no shutdown
Router(config)# end
Router# write memory
```

VERIFICATION COMMANDS

- Switch# show vlan brief
- Switch# show interfaces trunk
- Switch# show interfaces f0/1 switchport
- Switch# show interfaces f0/2 switchport
- Router# show ip interface brief
- Router# show run | section g0/0



TEST CONNECTIVITY

From PC1 (VLAN 10)

```
C:\> ping 192.168.20.10
```

Success ✓

From PC2 (VLAN 20)

```
C:\> ping 192.168.10.10
```

Success ✓

LEARNING OBJECTIVES

- ✓ Create VLANs on the switch
- ✓ Assign access ports to VLANs
- ✓ Configure trunk port (802.1Q)
- ✓ Configure router subinterfaces
- ✓ Enable inter-VLAN routing
- ✓ Verify connectivity between VLANs

NETWORK LAB #22

ADVANCED SUBNETTING LAB

Master subnet calculations using different prefix lengths for real-world networks.



LAB SCENARIO

You are assigned the network:

192.168.10.0/24

Your task is to create multiple smaller subnets for different departments.

ORIGINAL NETWORK

192.168.10.0/24

Subnet Mask:
255.255.255.0

Usable Hosts: **254**

SUBNETTING PROCESS

1. Start with Network Address
2. Select Prefix Length
3. Calculate Block Size
4. Find Network Address
5. Find Broadcast Address
6. Calculate Usable Host Range

DIVIDE INTO /26 SUBNETS

SUBNET	NETWORK ADDRESS	HOST RANGE	BROADCAST ADDRESS	USABLE HOSTS
1	192.168.10.0/26	192.168.10.1 – 192.168.10.62	192.168.10.63	62
2	192.168.10.64/26	192.168.10.65 – 192.168.10.126	192.168.10.127	62
3	192.168.10.128/26	192.168.10.129 – 192.168.10.190	192.168.10.191	62
4	192.168.10.192/26	192.168.10.193 – 192.168.10.254	192.168.10.255	62

COMMON PREFIX LENGTHS

PREFIX	SUBNET MASK	USABLE HOSTS
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14
/29	255.255.255.248	6



BLOCK SIZE FORMULA

Block Size = 256 – Last Octet of Subnet Mask

/25	→	256 – 128 =	128
/26	→	256 – 192 =	64
/27	→	256 – 224 =	32
/28	→	256 – 240 =	16
/29	→	256 – 248 =	8

Think in
BLOCKS!



Example /26 = Block Size 64
Subnets start at:
0, 64, 128, 192

WHAT YOU LEARN

- ✓ Advanced Subnetting
- ✓ Prefix Length Calculation
- ✓ Block Size Method
- ✓ Network & Broadcast Address
- ✓ Host Range Calculation



LAB CHALLENGE

Subnet 192.168.20.0/24 into /27 networks.
How many usable hosts are available in each subnet?



- A** 14 **B** 30 **C** 62 **D** 126

Comment your answer below!



NETWORK LABS



LEARN
Knowledge



PRACTICE
Hands-on Labs



MASTER
Networking Skills



CONNECT
The World

CIDR Notation & Subnet Masks

The table every CCNA student must memorize

CIDR	Subnet Mask	Usable Hosts	Network Bits	Host Bits
/24	255.255.255.0	254	24	8
/25	255.255.255.128	126	25	7
/26	255.255.255.192	62	26	6
/27	255.255.255.224	30	27	5
/28	255.255.255.240	14	28	4
/29	255.255.255.248	6	29	3
/30	255.255.255.252	2	30	2

Tip: /30 is used for point-to-point links
(only 2 hosts)

Example 3 | Advanced Level

You need at least 50 hosts per subnet from 192.168.1.0. Which subnet mask to use?

Step 1: We need 50 usable hosts

Step 2: Formula $\rightarrow 2^H - 2 \geq 50$

Step 3: Try $H=6 \rightarrow 2^6 - 2 = 62 \checkmark$ (62 is more than 50)

Step 4: So Host bits = 6, Network bits = $32 - 6 = 26$

Step 5: Answer $\rightarrow$ Use $/26 = 255.255.255.192$

Step 6: This gives 62 usable hosts per subnet $\checkmark$

Rule: Always pick the smallest subnet that fits your hosts. Don't waste IPs.

Example 2 | Medium Level

Given: 192.168.1.0/26 — Find everything

Step 1: /26 means 26 network bits, 6 host bits

Step 2: Subnet Mask $\rightarrow$ 255.255.255.192

Step 3: Block Size $\rightarrow 2^6 = 64$

Step 4: Subnets available $\rightarrow 2^2 = 4$ subnets

Step 5: Network Address $\rightarrow$ 192.168.1.0

Step 6: Broadcast $\rightarrow$ 192.168.1.63

Step 7: First Host $\rightarrow$ 192.168.1.1

Step 8: Last Host $\rightarrow$ 192.168.1.62

Step 9: Usable Hosts $\rightarrow 2^6 - 2 = 62$

Subnet ranges table

Subnet 1:	.0 to .63
Subnet 2:	.64 to .127
Subnet 3:	.128 to .191
Subnet 4:	.192 to .255

Example 1 | Easy Level

Given: 192.168.1.0/24 — Find everything

Step 1: Identify the class → Starts with 192
→ Class C

Step 2: Subnet Mask → /24 = 255.255.255.0

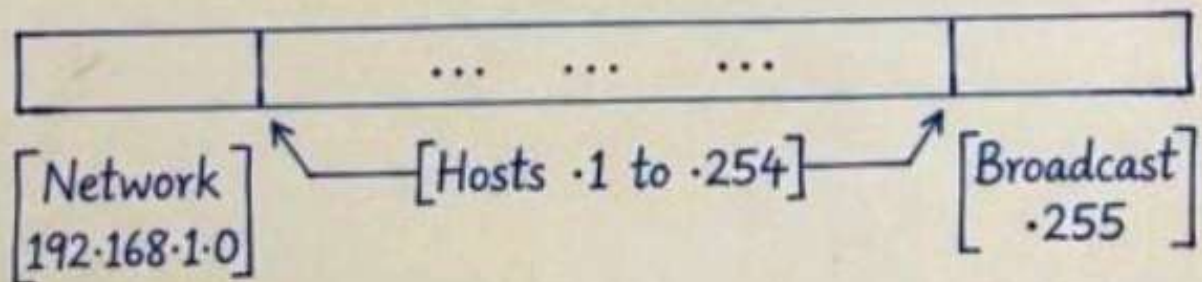
Step 3: Network Address → 192.168.1.0

Step 4: Broadcast Address → 192.168.1.255

Step 5: First Usable Host → 192.168.1.1

Step 6: Last Usable Host → 192.168.1.254

Step 7: Usable Hosts → $2^8 - 2 = 254$



IP Address Classes (Must Know)

Class	First Octet	Default Mask	Example	Use
A	1-126	255.0.0.0	10.0.0.1	Large networks
B	128-191	255.255.0.0	172.16.0.1	Medium networks
C	192-223	255.255.255.0	192.168.1.1	Small networks

Class D = Multicast | Class E = Experimental

Tip: Most home/office networks use Class C

Subnetting Cheat Sheet | Save This!

1) Key Formulas

- Usable Hosts = $2^H - 2$
- Number of Subnets = 2^S
- Block Size = 2^H

2) Steps to subnet any IP

1. Find the class (A/B/C)
2. Identify CIDR prefix
3. Calculate host bits ($32 - \text{prefix}$)
4. Find subnet mask
5. Calculate network, broadcast, first & last host

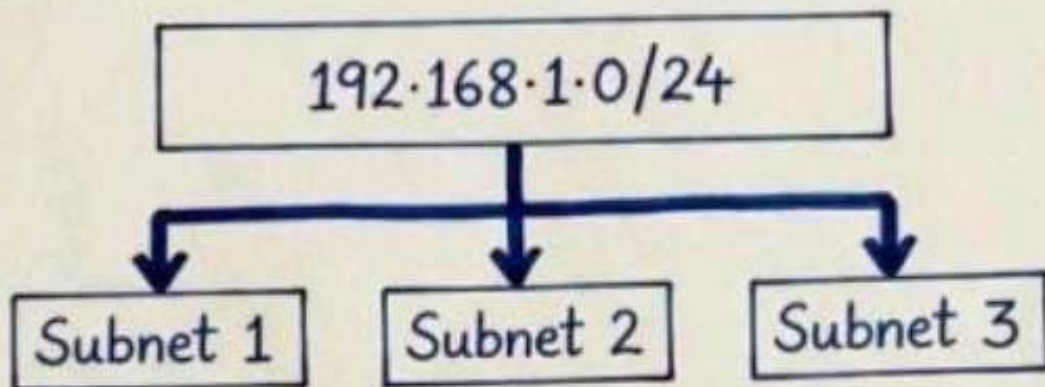
3) Common mistakes to avoid

- ✗ Don't forget to subtract 2 for usable hosts
- ✗ Network address is NOT a usable host
- ✗ Broadcast address is NOT a usable host
- ✗ $/32$ = single host, $/0$ = entire internet

Practice daily. Subnetting becomes easy in 1 week.

What is Subnetting?

- 1) Subnetting = Dividing one large network into smaller networks



- 2) Why do we subnet?

- Save IP addresses
- Better security
- Reduce network traffic
- Easier management

- 3) The Formula you MUST know:

$$\text{Usable Hosts} = 2^H - 2$$

$$\text{Subnets} = 2^S$$

where H = host bits

S = subnet bits

WHAT IS MPLS, RD, RT?

1 MPLS (Multiprotocol Label Switching)

MPLS is a high-performance network technology that forwards data using short labels instead of long IP addresses, enabling fast, scalable and reliable traffic engineering across service provider networks.



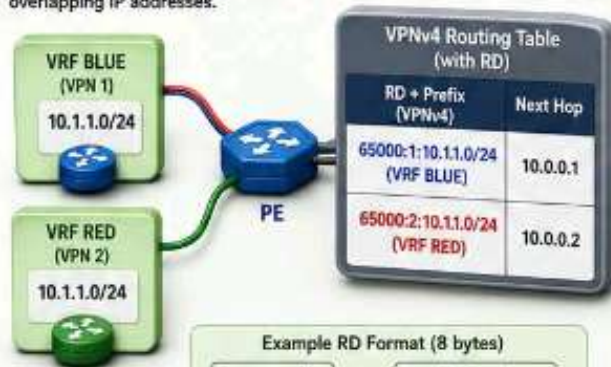
How MPLS Works

- 1 IP packet enters the MPLS network at Ingress LER.
- 2 Ingress LER adds a label to the packet and forwards it.
- 3 Each MPLS router forwards the packet using the label.
- 4 Egress LER removes the label and forwards the original IP packet.

- ✓ Faster forwarding (label lookup)
- ✓ Traffic Engineering (better control)
- ✓ Supports multiple services (VPN, VoIP, etc.)
- ✓ Scalable and provider grade

2 RD (Route Distinguisher)

RD is an 8-byte value added to VPNv4 routes to make them unique between different VPNs, even if they have overlapping IP addresses.



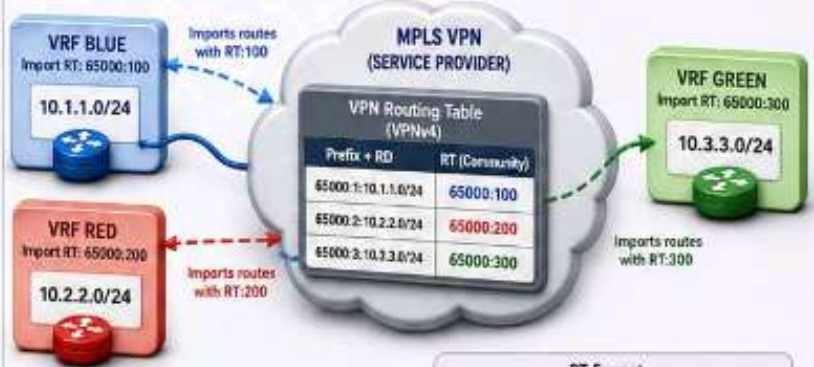
Example RD Format (8 bytes)

2 Bytes RD Type + 6 Bytes Administrator
Example: 65000:1 or 192.168.1.1:100

- ✓ RD does NOT control route import/export. It only makes routes unique.

3 RT (Route Target)

RT is a community attribute used to control which VPN routes are imported into a VRF. It defines the type of VPN membership.



Key Points

- ✓ RT controls route import/export between VPNs.
- ✓ Multiple VRFs can import the same RT (route sharing).
- ✓ A VRF can import multiple RTs.

RT Format

2 Bytes Type + 4 Bytes Administrator + 2 Bytes Assigned
Example: 65000:100



QUICK SUMMARY



MPLS

- Uses labels for fast forwarding
- Enables Traffic Engineering
- Basis for MPLS VPN services



RD

- Makes VPN routes unique
- Prevents overlapping IP conflicts
- Not used for import/export



RT

- Controls route import/export
- Defines VPN membership
- Uses community attribute



Together in MPLS VPN

- RD = Uniqueness
- RT = Policy (Import/Export)
- MPLS = Transport

WHAT'S INSIDE A FORTINET NETWORK RACK

— Every piece in place to protect, connect and power your business. —

1 FortiGate Firewall



Enterprise-grade firewall that inspects traffic, creates VPNs, and enforces security policies.

2 FortiSwitch



Manages network connectivity, users, devices and provides access to the network.

3 FortiAnalyzer



Collects and analyzes logs from all devices and generates reports and insights.

4 FortiManager



Centralized management for all Fortinet devices, policies, updates and monitoring.

5 Patch Panel



Organizes and manages RJ45 network cables in a structured manner.

6 Cable Manager



Keeps cables organized and properly routed inside the rack.

7 Fiber Patch Panel



Manages fiber optic cables for high-speed and long-distance connectivity.

8 Core Switch



High-speed Layer 3 switch that handles routing, VLANs and advanced network traffic.

9 UPS



Provides backup power to keep devices running during power outages.

10 Rack PDU



Distributes power to all devices in the rack safely and efficiently.

11 NAS Storage



Network-attached storage for secure backup, file sharing, and centralized data storage.

12 Servers



Hosts applications and services such as Active Directory, DNS, DHCP, File Server and more.



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Security
& Protection



Networking
& Connectivity



Devices
& Infrastructure



Storage
& Backup



Power
& Energy



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SUBNETTING

Divide One Network into Multiple Smaller Networks



1 WHAT IS SUBNETTING?

Subnetting is the process of dividing a large network into smaller, manageable networks called subnets.

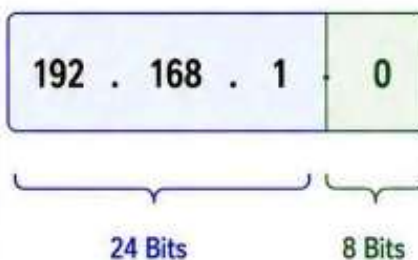
- ✓ Better Network Performance
- ✓ Reduced Broadcast Traffic
- ✓ Improved Security
- ✓ Efficient IP Address Utilization



2 IPV4 ADDRESS STRUCTURE

192.168.1.0 /24

Network Portion Host Portion



3 COMMON CIDR PREFIXES

CIDR	SUBNET MASK	USABLE HOSTS
/24	255.255.255.0	254
/25	255.255.255.128	126
/26	255.255.255.192	62
/27	255.255.255.224	30
/28	255.255.255.240	14

4 SUBNETTING EXAMPLE

Original Network

192.168.1.0/24

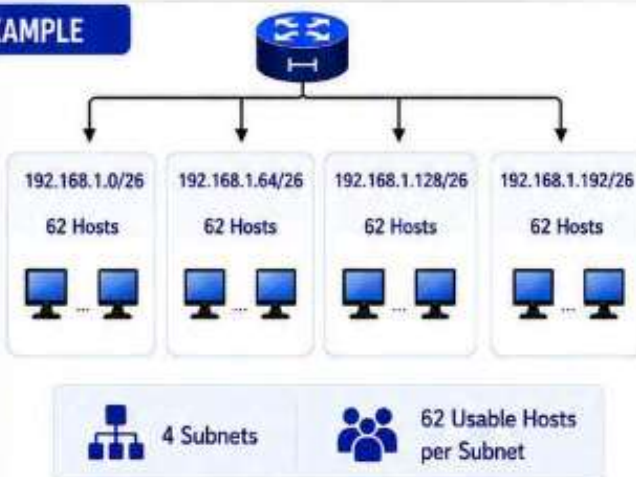
Divide into /26 Subnets

192.168.1.0/26

192.168.1.64/26

192.168.1.128/26

192.168.1.192/26



5 SUBNETTING FORMULA

Number of Subnets

$$2^n$$

Where:

n = Borrowed Bits

Usable Hosts per Subnet

$$2^h - 2$$

h = Host Bits

6 SUBNETTING TERMS

Network Address First IP in the subnet	192.168.1.0
First Usable Host	192.168.1.1
Last Usable Host	192.168.1.62
Broadcast Address Last IP in the subnet	192.168.1.63

7 FAST SUBNETTING TRICK

CIDR	BLOCK SIZE
/25	128
/26	64
/27	32
/28	16
/29	8
/30	4



TIP

Block size is the increment between two subnets in the last octet.

Example:
/26 (64) means
0, 64, 128, 192

8 REAL-WORLD USE CASE



NETWORKING WITH ISRAR
LEARN | PRACTICE | SUCCEED





Q What is the maximum number of hosts in a **/24** network?



A 254

B 255

C 256

D 512



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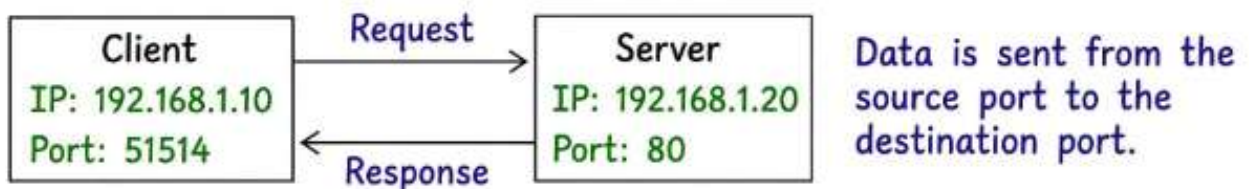


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Ports in Networking

What is a Port?

A port is a logical endpoint (a number between 0 and 65535) used to identify a specific process or service running on a host. It helps to deliver data to the correct application on a device. It works with IP address to enable communication.



Types of Ports

Sun Networking Academy

1. Well-Known Ports (0 - 1023)

- Reserved for system services and common protocols.
- Examples: 20 (FTP-Data), 21 (FTP-Control), 22 (SSH), 80 (HTTP), 443 (HTTPS), 53 (DNS).

2. Registered Ports (1024 - 49151)

- Registered with IANA for specific applications and services.
- Used by user applications and some-system processes.
- Examples: 3306 (MySQL), 1433 (MS SQL Server), 3389 (RDP).

3. Dynamic / Private / Ephemeral Ports (49152 - 65535)

- Used temporarily by client applications.
- Assigned dynamically by the OS when a connection is initiated.
- Also called ephemeral ports.

Key Points

- Each port is associated with a specific service on a host.
- Communication is identified by a pair: **Source IP : Source Port**
Destination IP : Destination Port
- Example: A web request --> 192.168.1.10:51514 --> 192.168.1.20:80

What is VLAN TAGGING?

VLAN Tagging (IEEE 802.1Q)

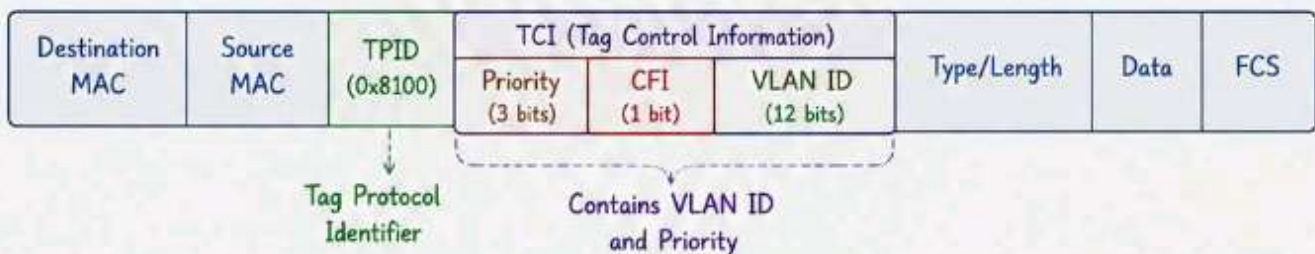
Key Points:

- VLAN Tagging adds VLAN information to Ethernet frames.
- Enables multiple VLANs to traverse the same physical link.
- Uses IEEE 802.1Q standard to insert a 4-byte tag.

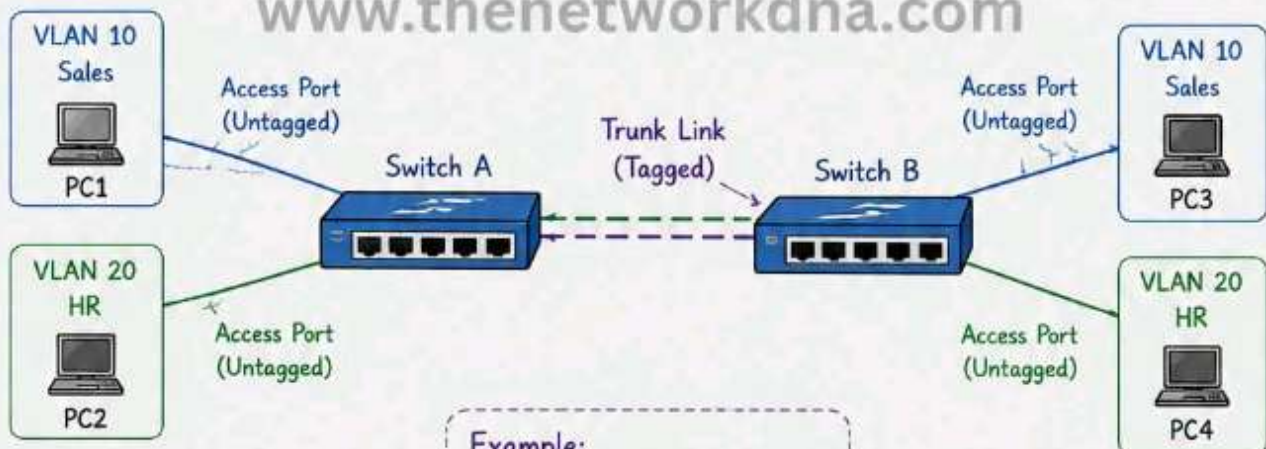
How It Works:

- Frames are tagged on **Trunk** ports when leaving the switch.
- The tag includes **VLAN ID**, which identifies the VLAN.
- Frames are untagged on **Access** ports for devices in the same VLAN.

Frame Format (802.1Q Tag):



www.thenetworkdna.com



Tagging Rules:

- Access Port:
 - Sends frames **UNTAGGED**
 - Belongs to one VLAN only
- Trunk Port:
 - Sends frames **TAGGED**
 - Carries multiple VLANs

Example:

If a frame from PC1 (VLAN 10) goes to PC3 (VLAN 10) through a trunk link:

1. PC1 sends untagged frame to Switch A
2. Switch A adds **VLAN 10** tag to the frame
3. Frame travels across the trunk link (tagged)
4. Switch B removes the tag and sends untagged frame to PC3

Benefits:

- ✓ Improves Security by separating traffic
- 🕒 Efficient Use of Bandwidth with multiple VLANs on one link
- 🔗 Scalable and Easy to Manage



Quick Reminder:

Tag on **Trunk**, Untag on **Access**!



Standard:

IEEE 802.1Q



CHOOSE THE RIGHT TRANSCEIVER



SPEED

From 1G to 800G+
Choose the speed you need.



DISTANCE

Copper for short distance, Fiber for long distance.



INTERFACE

SFP, SFP+, QSFP+, QSFP28, QSFP56, QSFP-DD, OSFP.



MEDIA

MMF, SMF, CWDM, DWDM, DAC, AOC.



APPLICATION

Data centers, Enterprise networks, Service providers & more.

TYPE	1G	10G	25G	40G	100G	200G	400G	800G	DAC	AOC	RJ45	CWDM/DWDM
FORM FACTOR	SFP	SFP+	SFP28	QSFP+	QSFP28	QSFP56	QSFP-DD	OSFP	SFP/QSFP+	SFP/QSFP+	SFP	SFP/SFP+
MAX SPEED	1.25 Gbps	10.3 Gbps	25.78 Gbps	40 Gbps	100 Gbps	200 Gbps	400 Gbps	800 Gbps	10-400 Gbps	10-400 Gbps	10 Gbps	10-100 Gbps
CABLE TYPE	Fiber	Fiber	Fiber	Fiber	Fiber	Fiber	Fiber	Fiber	Copper	Fiber	Copper	Fiber
USE CASE	Access Networks	Enterprise Networks	High Density Networks	Data Center & Cloud	Data Center Core	Next-Gen Data Center	AI/ML Networks	Ultra High Speed Networks	Short Range Links	Long Range Links	Copper Networks	Metro & Long Haul Networks



HIGH PERFORMANCE
Built for speed and reliability



FLEXIBLE OPTIONS
Multiple form factors and media types



EXTENDED REACH
From short-range to long-haul links



HOT SWAPPABLE
Plug and play convenience



WIDE COMPATIBILITY
Works with major network equipment

Datacenter Rack-4

Patch Panels



Firewall

NAS

Router

Switch

Server Backup

UPS

Servers



PDU

IPSec TUNNEL

BETWEEN TWO FORTINET FIREWALLS



FORTIGATE-1 CONFIGURATION

1. CREATE IPSEC PHASE 1 (IKE)

```
config vpn ipsec phase1-interface
edit "IPSEC_TO_FG2"
set interface "port1"
set peertype any
set remote-gw 203.0.113.2
set ike-version 2
set proposal aes256-sha256
set dhgrp 14
set psksecret "YourPreSharedKey123"
next
```

2. CREATE IPSEC PHASE 2 (SELECTOR)

```
config vpn ipsec phase2-interface
edit "IPSEC_TO_FG2"
set phase1name "IPSEC_TO_FG2"
set proposal aes256-sha256
set pfs enable
set dhgrp 14
set src-subnet 10.10.1.0 255.255.255.0
set dst-subnet 10.20.1.0 255.255.255.0
next
```

3. CREATE POLICY & ALLOW TRAFFIC

```
config firewall policy
edit 10
set name "LAN_TO_LAN_IPSEC"
set srcintf "port2"
set dstintf "IPSEC_TO_FG2"
set srcaddr "LAN1_NET"
set dstaddr "LAN2_NET"
set action accept
set schedule "always"
set service "ALL"
set nat disable
next
```

FORTIGATE-2 CONFIGURATION

1. CREATE IPSEC PHASE 1 (IKE)

```
config vpn ipsec phase1-interface
edit "IPSEC_TO_FG1"
set interface "port1"
set peertype any
set remote-gw 203.0.113.1
set ike-version 2
set proposal aes256-sha256
set dhgrp 14
set psksecret "YourPreSharedKey123"
next
```

2. CREATE IPSEC PHASE 2 (SELECTOR)

```
config vpn ipsec phase2-interface
edit "IPSEC_TO_FG1"
set phase1name "IPSEC_TO_FG1"
set proposal aes256-sha256
set pfs enable
set dhgrp 14
set src-subnet 10.20.1.0 255.255.255.0
set dst-subnet 10.10.1.0 255.255.255.0
next
```

3. CREATE POLICY & ALLOW TRAFFIC

```
config firewall policy
edit 10
set name "LAN_TO_LAN_IPSEC"
set srcintf "port2"
set dstintf "IPSEC_TO_FG1"
set srcaddr "LAN2_NET"
set dstaddr "LAN1_NET"
set action accept
set schedule "always"
set service "ALL"
set nat disable
next
```

IPSEC TUNNEL OVERVIEW

- ✓ IPsec encrypts traffic between two FortiGate firewalls.
- ✓ Provides confidentiality, integrity & authentication.
- ✓ Uses IKE (Phase 1) to establish the secure tunnel.
- ✓ Uses IPsec (Phase 2) to protect data traffic.
- ✓ Supports site-to-site and remote access VPNs.

KEY SETTINGS

- IKE Version : 2
- Encryption : AES-256
- Authentication : SHA-256
- DH Group : 14
- Key Life : 28800 sec (default)
- PFS : Enabled

EXAMPLE TOPOLOGY SUMMARY

- FortiGate-1 Public IP : 203.0.113.1
- FortiGate-2 Public IP : 203.0.113.2
- LAN 1 (FG-1) : 10.10.1.0/24
- LAN 1 (FG-2) : 10.20.1.0/24
- Tunnel Interface : IPSEC_TO_FG1 / IPSEC_TO_FG2
- Pre-Shared Key : YourPreSharedKey123

VERIFICATION COMMANDS

```
get vpn ipsec tunnel summary
get vpn ipsec sa
execute ping-options source <lan-ip>
execute ping <remote-lan-ip>
diagnose vpn ike gateway list
diagnose sniffer packet any "esp" 4
```

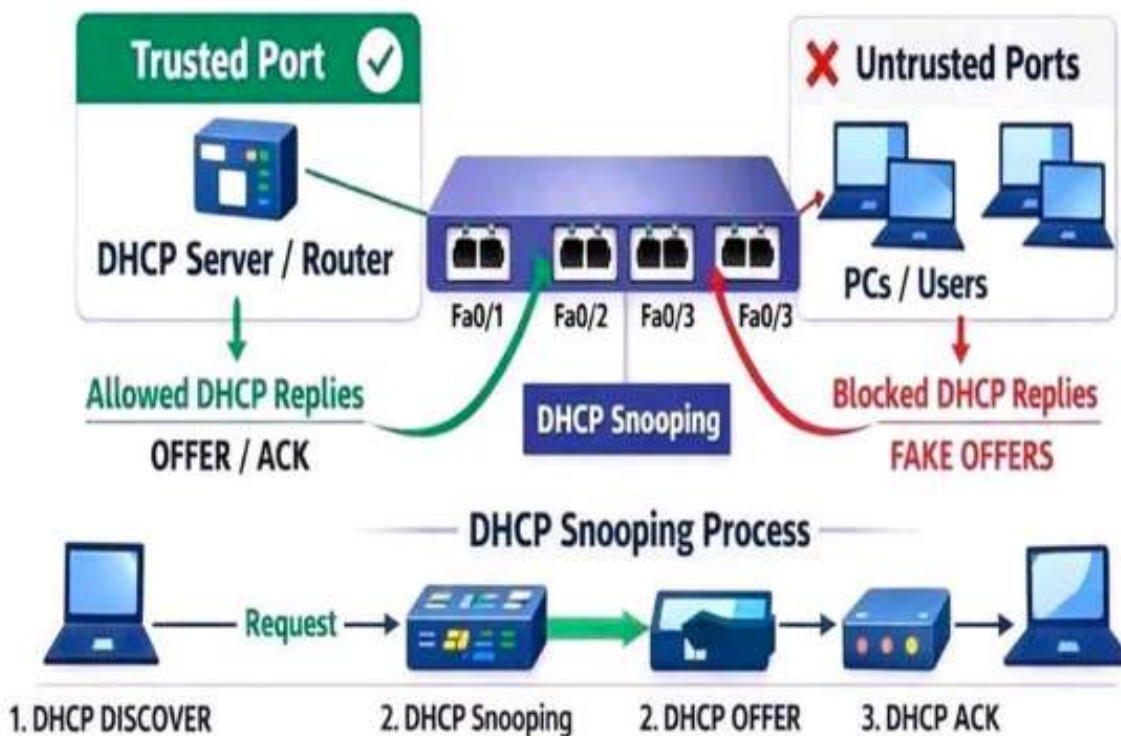
TROUBLESHOOTING TIPS

- Ensure public IPs are reachable.
- Check firewall policies & NAT settings.
- Verify Phase 1 & Phase 2 are up.
- Check proposals, PFS & traffic selectors.
- Use diagnose commands for debugging.

SUCCESS CHECKLIST

- ✓ Phase 1 (IKE) and Phase 2 (IPsec) are UP.
- ✓ Ping between LANs is successful.
- ✓ Traffic is passing through the encrypted tunnel.
- ✓ Monitor with: get vpn ipsec tunnel summary

DHCP Snooping



DHCP Snooping Binding Table

MAC Address	IP Address	VLAN	Port
00:1A:2B:3C:4D:5E	192.168.1.10	10	Fa0/2

DHCP Snooping Commands

1. Enable DHCP Snooping globally:
 - Switch(config)# ip dhcp snooping
 - Switch(config)# ip dhcp snooping vlan 10
2. Set the trusted interface:
 - Switch(config)# interface fa0/1
 - Switch(config-if)# ip dhcp snooping trust
3. Rate limit untrusted interface:
 - Switch(config)# ip dhcp snooping limit rate 5
4. Disable Option 82:
 - no ip dhcp snooping information option

JS

script.js

```
1 // Expense Tracker - JavaScript
2 document.addEventListener("DOMContentLoaded", () => {
3   // Elements
4   const descriptionInput = document.getElementById("description");
5   const amountInput = document.getElementById("amount");
6   const categoryInput = document.getElementById("category");
7   const addButton = document.getElementById("add-expense");
8   const totalAmountEl = document.getElementById("total-amount");
9   const monthlyAmountEl = document.getElementById("Monthly-amount");
10  const tbody = document.getElementById("expense-body");
11
12  let expenses = [];
13
14  // Add Expense
15  addButton.addEventListener("click", addExpense);
16
17  function addExpense() {
18    const description = descriptionInput.value.trim();
19    const amount = parseFloat(amountInput.value);
20    const category = categoryInput.value;
21
22    if (!description || isNaN(amount) || amount <= 0) {
23      alert("Please fill in all fields correctly.");
24      return;
25    }
26
27    const expense = {
28      id: Date.now(),
29      description,
30      amount,
31      category,
32      date: new Date().toLocaleDateString()
33    };
34
35    expenses.push(expense); // Add to list
36    renderExpenses();
37    updateSummary();
38    clearInputs();
39  }
40
41  // Render Expenses Table
42  function renderExpenses() {
43    tbody.innerHTML = "";
44    expenses.forEach(expense => {
45      const row = document.createElement("tr");
46      row.innerHTML = `
47        <td>${expense.description}</td>
48        <td>${expense.category}</td>
49        <td>${expense.amount.toFixed(2)}</td>
50        <td>${expense.date}</td>
51        <td>
52          <button class="delete-btn" data-id="${expense.id}">✖</button>
53        </td>
54      `;
55      tbody.appendChild(row);
56    });
57  }
```

Expense Tracker

Track your expenses. Build a better tomorrow.

Description

e.g., Groceries

Amount

0.00

Category

Select Category

+ Add Expense



Total Expenses

\$668.50



This Month

\$528.50

Recent Expenses

[View All](#)

Description	Category	Amount	Date	Action
Groceries	Food	\$45.50	May 18, 2024	
Bus Pass	Transport	\$60.00	May 17, 2024	
Electricity Bill	Utilities	\$120.00	May 16, 2024	
Headphones	Entertainment	\$80.00	May 14, 2024	
Movie Ticket	Entertainment	\$15.00	May 14, 2024	
Dinner	Food	\$48.00	May 13, 2024	

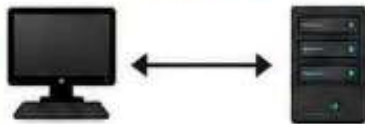
Showing 1 to 6 of 6 expenses

PACKET vs FRAME vs SEGMENT

DATA UNITS IN THE NETWORK STACK

SEGMENT

Data unit at the
Transport Layer
(Layer 4)



Created by:
TCP / UDP

PURPOSE

End-to-end communication
between applications

ADDRESSES USED

Port Numbers
(Source / Destination)

PDU STRUCTURE



Segment

EXAMPLE

Web browsing, File transfer,
Email, Video streaming

PACKET

Data unit at the
Network Layer
(Layer 3)



Created by:
IP (IPv4 / IPv6)

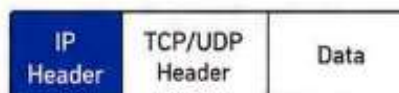
PURPOSE

Logical addressing and
routing across networks

ADDRESSES USED

IP Address
(Source / Destination)

PDU STRUCTURE



Packet

EXAMPLE

IP packet traveling from
one network to another

FRAME

Data unit at the
Data Link Layer
(Layer 2)



Created by:
Ethernet / PPP / HDLC

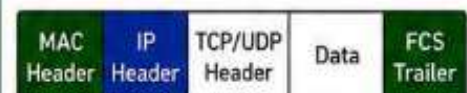
PURPOSE

Physical addressing and
delivery on local network

ADDRESSES USED

MAC Address
(Source / Destination)

PDU STRUCTURE



Frame

EXAMPLE

Ethernet frame on a
local network

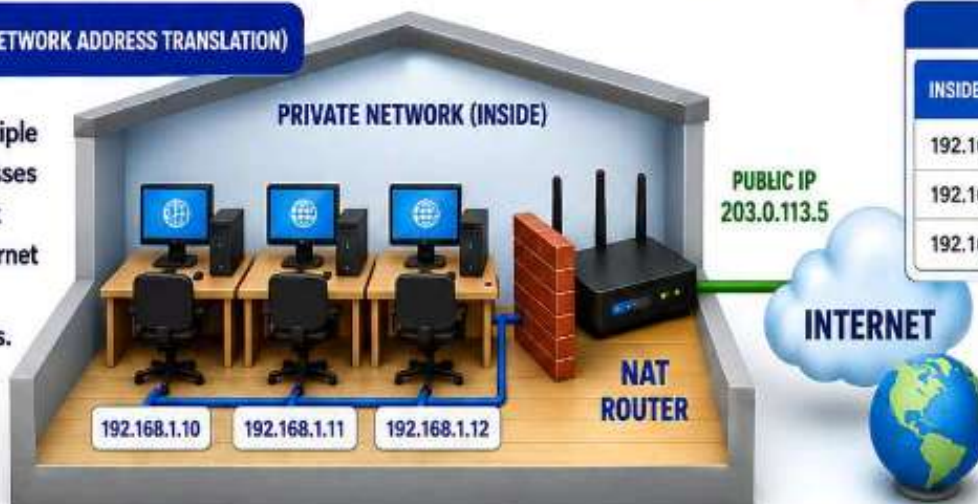
SUMMARY

- SEGMENT** → Used by Transport Layer (TCP/UDP) to move data between applications.
- PACKET** → Used by Network Layer (IP) to move data between networks.
- FRAME** → Used by Data Link Layer (Ethernet) to move data between devices on the same network.

WHAT IS NAT AND PAT?

1 NAT (NETWORK ADDRESS TRANSLATION)

NAT allows multiple private IP addresses inside a network to access the internet using a single public IP address.



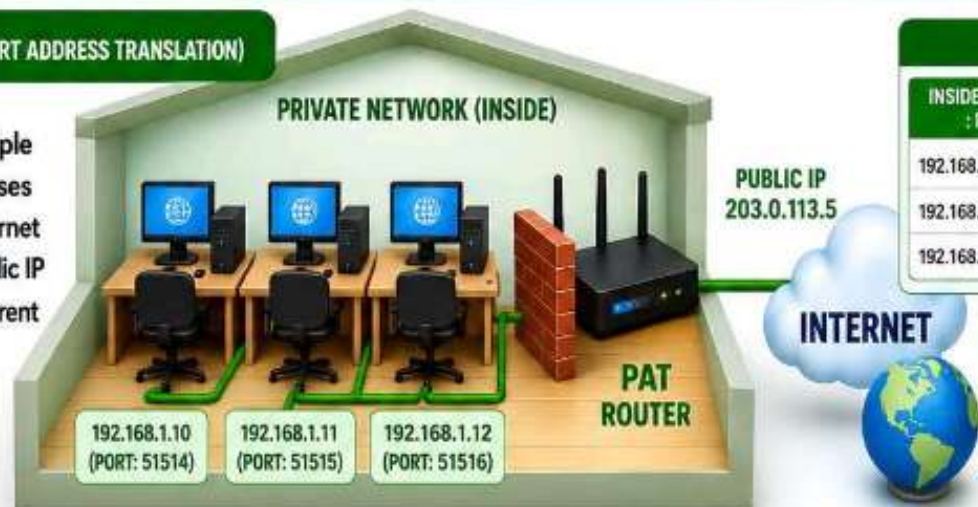
HOW NAT WORKS

INSIDE LOCAL IP	TRANSLATED TO PUBLIC IP
192.168.1.10	→ 203.0.113.5
192.168.1.11	→ 203.0.113.5
192.168.1.12	→ 203.0.113.5

- ✓ All devices share the same public IP address. The router keeps track of connections and sends responses back to the correct internal device.

2 PAT (PORT ADDRESS TRANSLATION)

PAT allows multiple private IP addresses to access the internet using a single public IP address with different port numbers.



HOW PAT WORKS

INSIDE LOCAL IP : PORT	TRANSLATED TO PUBLIC IP : PORT
192.168.1.10:51514	→ 203.0.113.5:10001
192.168.1.11:51515	→ 203.0.113.5:10002
192.168.1.12:51516	→ 203.0.113.5:10003

- ✓ The router translates private IP addresses and assigns unique port numbers on the public IP. This allows many more devices to share a single public IP.



WHY USE NAT/PAT?

- ✓ Conserves public IP addresses
- ✓ Adds a layer of security
- ✓ Hides internal network structure

KEY DIFFERENCE

NAT maps IP to IP (one to one / many to one)

PAT maps IP + Port to IP + Port (many to one using different ports)



ANALOGY

NAT: Many people using one car

PAT: Many people using one car with different seat numbers



COMMON USE

Used in home routers, enterprise gateways, and cloud networks.



SEND EMAIL USING PHPMailer

Easy Example for Beginners

PHPMailer is a popular PHP library that helps you send emails easily using SMTP. No need to use the PHP mail() function—just configure and send!

Simple
Powerful
Reliable

1 INSTALL PHPMailer

- Download PHPMailer from github.com/PHPMailer/PHPMailer
- Extract the ZIP file
- Upload the PHPMailer folder to your project
- Include `autoload.php`



2 CONFIGURE SMTP

- You need SMTP details from your email provider (Gmail, Outlook, etc.)
- Use secure SMTP settings (TLS/465 or SSL/587)
- Update the code with your credentials



3 SEND EMAIL

- Create PHPMailer object
- Set sender, recipient and content
- Call `send()`



1 Basic Email Sending

```
<?php
use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\Exception;

require 'vendor/autoload.php';

$mail = new PHPMailer(true);

try {
    $mail->isSMTP();
    $mail->Host = 'smtp.gmail.com';
    $mail->SMTPAuth = true;
    $mail->Username = 'your@gmail.com';
    $mail->Password = 'your_app_password';
    $mail->SMTPSecure = 'tls';
    $mail->Port = 587;

    $mail->setFrom('you@gmail.com', 'My Site');
    $mail->addAddress('friend@example.com');
    $mail->Subject = 'Hello from PHPMailer';
    $mail->Body = 'This is a simple test';
    $mail->send();
    echo 'Email sent successfully!';
} catch (Exception $e) {
    echo 'Error: ' . $mail->ErrorInfo;
}
```

2 Sending HTML Email

```
<?php
use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\Exception;

require 'vendor/autoload.php';

$mail = new PHPMailer(true);

try {
    $mail->isSMTP();
    $mail->Host = 'smtp.gmail.com';
    $mail->SMTPAuth = true;
    $mail->Username = 'your@gmail.com';
    $mail->Password = 'your_app_password';
    $mail->SMTPSecure = 'tls';
    $mail->Port = 587;

    $mail->setFrom('you@gmail.com', 'My Site');
    $mail->addAddress('friend@example.com');
    $mail->Subject = 'HTML Email Example';
    $mail->isHTML(true);
    $mail->Body = 'Hi! Hello!<br>This is an email<br>with formatting, to HTML!';
    $mail->send();
} catch (Exception $e) {
    echo 'Error: ' . $mail->ErrorInfo;
}
```

3 Adding Attachments

```
<?php
use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\Exception;

require 'vendor/autoload.php';

$mail = new PHPMailer(true);

try {
    $mail->isSMTP();
    $mail->Host = 'smtp.gmail.com';
    $mail->SMTPAuth = true;
    $mail->Username = 'your@gmail.com';
    $mail->Password = 'your_app_password';
    $mail->SMTPSecure = 'tls';
    $mail->Port = 587;

    $mail->setFrom('you@gmail.com', 'My Site');
    $mail->addAddress('friend@example.com');
    $mail->Subject = 'Report with Attachment';
    $mail->Body = 'Please find the attachment';
    $mail->addAttachment('report.pdf');

    $mail->send();
} catch (Exception $e) {
    echo 'Error: ' . $mail->ErrorInfo;
}
```

4 Using CC and BCC

```
<?php
use PHPMailer\PHPMailer\PHPMailer;
use PHPMailer\PHPMailer\Exception;

require 'vendor/autoload.php';

$mail = new PHPMailer(true);

try {
    $mail->isSMTP();
    $mail->Host = 'smtp.gmail.com';
    $mail->SMTPAuth = true;
    $mail->Username = 'your@gmail.com';
    $mail->Password = 'your_app_password';
    $mail->SMTPSecure = 'tls';
    $mail->Port = 587;

    $mail->setFrom('you@gmail.com', 'My Site');
    $mail->addAddress('friend@example.com');
    $mail->addCC('cc@example.com'); // CC
    $mail->addBCC('bcc@example.com'); // BCC
    $mail->Subject = 'Meeting Update';
    $mail->Body = 'Please find the latest update.';

    $mail->send();
} catch (Exception $e) {
    echo 'Error: ' . $mail->ErrorInfo;
}
```

Common PHPMailer Methods

Method	Description
<code>isSMTP()</code>	Tells PHPMailer to use SMTP
<code>setFrom(\$email, \$name)</code>	Sets the sender email and name
<code>addAddress(\$email, \$name)</code>	Adds a primary recipient
<code>addCC(\$email, \$name)</code>	Adds a CC recipient
<code>addBCC(\$email, \$name)</code>	Adds a BCC recipient
<code>Subject(\$subject)</code>	Sets the email subject
<code>Body(\$body)</code>	Sets the email body (plain text or HTML)
<code>addAttachment(\$path)</code>	Adds a file attachment
<code>send()</code>	Sends the email

SMTP Settings (Example - Gmail)

- ✓ Host: `smtp.gmail.com`
- ✓ Port: 587 (TLS) or 465 (SSL)
- ✓ Encryption: TLS / SSL
- ✓ SMTP Auth: Yes
- ✓ Username: your Gmail address
- ✓ Password: App Password (not your main Gmail password) for OAuth

For Gmail, enable 2-Step Verification and create an App Password.

How It Works

- 1 You install PHPMailer and include it in your project.
- 2 You configure SMTP settings with your email provider.
- 3 You create a PHPMailer object.
- 4 You set sender, recipients (To, CC, BCC), subject and body.
- 5 You call `send()` to send the email.
- 6 PHPMailer connects to the SMTP server and delivers the email.

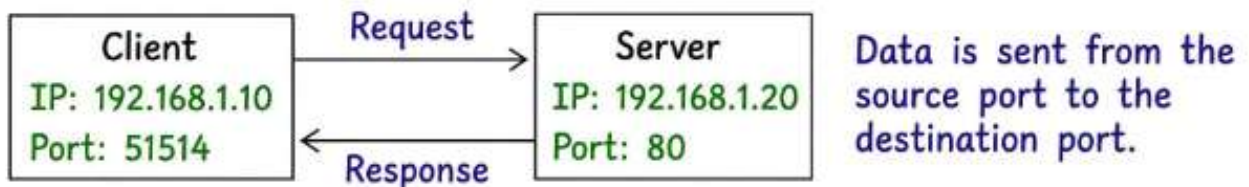
That's it! You are ready to send emails like a pro! 🌟



Ports in Networking

What is a Port?

A port is a logical endpoint (a number between 0 and 65535) used to identify a specific process or service running on a host. It helps to deliver data to the correct application on a device. It works with IP address to enable communication.



Types of Ports

Sun Networking Academy

1. Well-Known Ports (0 - 1023)

- Reserved for system services and common protocols.
- Examples: 20 (FTP-Data), 21 (FTP-Control), 22 (SSH), 80 (HTTP), 443 (HTTPS), 53 (DNS).

2. Registered Ports (1024 - 49151)

- Registered with IANA for specific applications and services.
- Used by user applications and some-system processes.
- Examples: 3306 (MySQL), 1433 (MS SQL Server), 3389 (RDP).

3. Dynamic / Private / Ephemeral Ports (49152 - 65535)

- Used temporarily by client applications.
- Assigned dynamically by the OS when a connection is initiated.
- Also called ephemeral ports.

Key Points

- Each port is associated with a specific service on a host.
- Communication is identified by a pair: **Source IP : Source Port**
Destination IP : Destination Port
- Example: A web request --> 192.168.1.10:51514 --> 192.168.1.20:80

Q. WHY WE NEED IP ADDRESS WHEN WE ALREADY HAD MAC ADDRESS?

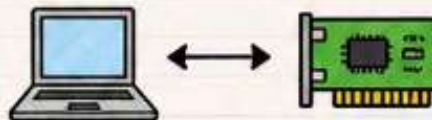


ANSWER: Both MAC Address and IP Address are used in networking, but they work at **different layers** and serve **different purposes**.

1) WHAT IS MAC ADDRESS ?

- MAC (Media Access Control) address is a **unique physical address** assigned to a **Network Interface Card (NIC)** by the manufacturer.
- It works at **Data Link Layer (Layer 2)**
- **Format:** 48-bit (e.g. **00:1A:2B:3C:4D:5E**)

MAC Address (Layer 2)



Example: **00:1A:2B:3C:4D:5E**

Key Points

- ✓ **Hardcoded** in the hardware
- ✓ **Cannot be changed**
- ✓ Works only within the **local network**



2) WHAT IS IP ADDRESS ?

- IP (Internet Protocol) address is a **logical address** assigned to a device for communication **across networks**.
- It works at **Network Layer (Layer 3)**
- **Format:** 32-bit IPv4 (e.g. **192.168.1.10**)
128-bit IPv6 (e.g. **2001:db8::1**)

IP Address (Layer 3)



Example: **192.168.1.10**

Key Points

- ✓ Assigned by **OS / DHCP / Manually**
- ✓ **Can be changed**
- ✓ Works **across different networks**



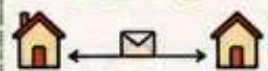
3) KEY DIFFERENCES

Feature	MAC Address (Layer 2)	IP Address (Layer 3)
Purpose	Used for communication in local network	Used for communication across different networks
Address Type	Physical Address	Logical Address
Length	48-bit	32-bit (IPv4) / 128-bit (IPv6)
Assigned By	Manufacturer (Burned in NIC)	Network Admin / DHCP / ISP
Changeable	No	Yes
Example	00:1A:2B:3C:4D:5E	192.168.1.10

4) WHY BOTH ARE NEEDED ?

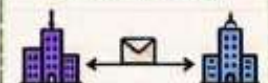
- MAC address helps in delivering data within the **local network**.
- IP address helps in delivering data to **remote networks**.
- Think of it like this:
MAC = Home Address (Delivery inside locality)
IP = City Address (Delivery to another city)

Same City



MAC Address

Different City



IP Address

5) SUMMARY

MAC address gets the data to the next device in the **local network**,
IP address gets the data to the correct destination **anywhere in the world**.



They **work together** to make the **Internet possible** !



Remember:

Layers are different,
Functions are different,
But Goal is the same -
Communication!

OSI MODEL Mindmap



OSPF MULTI-AREA CONFIGURATION LAB

DESIGN AND CONFIGURE OSPF IN A MULTI-AREA ENVIRONMENT

OBJECTIVE

Configure OSPF in a multi-area environment to achieve scalability, efficient routing and proper area backbone communication.

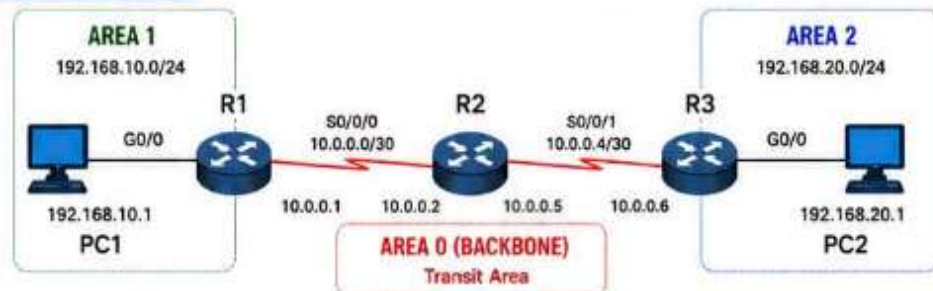


TASKS

1. Configure IP addresses on all routers.
2. Enable OSPF process on all routers.
3. Configure areas (Area 0 backbone, Area 1 and Area 2).
4. Advertise networks in the correct areas.
5. Verify adjacency and routing.
6. Test end-to-end connectivity.



TOPOLOGY



IP ADDRESSING TABLE

Device	Interface	IP Address	Subnet Mask	Area	Description
R1	G0/0	192.168.10.1	255.255.255.0	1	LAN 1 Gateway
R1	S0/0/0	10.0.0.1	255.255.255.252	0	To R2
R2	S0/0/0	10.0.0.2	255.255.255.252	0	To R1
R2	S0/0/1	10.0.0.5	255.255.255.252	0	To R3
R3	S0/0/1	10.0.0.6	255.255.255.252	0	To R2
R3	G0/0	192.168.20.1	255.255.255.0	2	LAN 2 Gateway
PC1	NIC	192.168.10.10	255.255.255.0	-	GW: 192.168.10.1
PC2	NIC	192.168.20.10	255.255.255.0	-	GW: 192.168.20.1

R1 - OSPF CONFIGURATION (AREA 1)

```
1 enable
2 configure terminal
3 hostname R1
4 interface g0/0
5 ip address 192.168.10.1 255.255.255.0
6 no shutdown
7 interface s0/0/0
8 ip address 10.0.0.1 255.255.255.252
9 no shutdown
10 router ospf 1
11 router-id 1.1.1.1
12 network 192.168.10.0 0.0.0.255 area 1
13 network 10.0.0.0 0.0.0.3 area 0
```



R2 - OSPF CONFIGURATION (AREA 0)

```
1 enable
2 configure terminal
3 hostname R2
4 interface s0/0/0
5 ip address 10.0.0.2 255.255.255.252
6 no shutdown
7 interface s0/0/1
8 ip address 10.0.0.5 255.255.255.252
9 no shutdown
10 router ospf 1
11 router-id 2.2.2.2
12 network 10.0.0.0 0.0.0.3 area 0
13 network 10.0.0.4 0.0.0.3 area 0
```



R3 - OSPF CONFIGURATION (AREA 2)

```
1 enable
2 configure terminal
3 hostname R3
4 interface s0/0/1
5 ip address 10.0.0.6 255.255.255.252
6 no shutdown
7 interface g0/0
8 ip address 192.168.20.1 255.255.255.0
9 no shutdown
10 router ospf 1
11 router-id 3.3.3.3
12 network 10.0.0.4 0.0.0.3 area 0
13 network 192.168.20.0 0.0.0.255 area 2
```



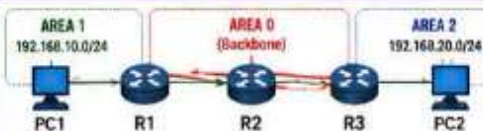
VERIFICATION COMMANDS

- show ip ospf neighbor
- show ip ospf interface brief
- show ip route ospf
- show ip protocols
- ping 192.168.20.10 (from PC1)
- ping 192.168.10.10 (from PC2)
- traceroute 192.168.20.10 (from PC1)



These commands help you verify adjacency, routing and end-to-end connectivity.

HOW IT WORKS



1. R1 belongs to Area 1 and connects to backbone (Area 0).
2. R2 is the backbone router connecting Area 1 and Area 2.
3. R3 belongs to Area 2 and connects to backbone (Area 0).
4. All inter-area traffic goes through Area 0.
5. OSPF calculates the best path and maintains routing tables.

★ Area 0 is the backbone of OSPF and is required.

LEARNING POINTS

- OSPF supports hierarchical routing using areas.
- Area 0 is the backbone and must connect all areas.
- ABR (Area Border Router) connects Area 0 with other areas.
- LSAs are not propagated between areas, improving scalability.
- Use router-id command for stable OSPF ID.
- Always verify adjacency and routing after configuration.

CHALLENGE QUESTION

If the link between R2 and R3 goes down, how will OSPF react in this topology? Will PC1 (192.168.10.10) be able to reach PC2 (192.168.20.10)? Why or why not? What will be the change in routing tables?



Comment your answer below! 🗨️

QUICK TEST



Which area is the backbone area?



Can Area 1 and Area 2 communicate directly?



What happens if Area 0 is not connected?



Which command shows OSPF neighbors?



Will inter-area routes be propagated?





NETWORKING
WITH ISRAR

LEARN | PRACTICE | SUCCEED

SUBNETTING

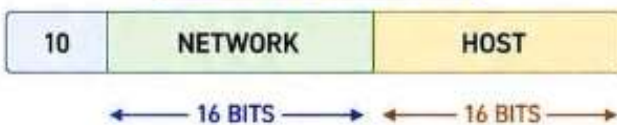
IPV4 SUBNETTING MADE EASY



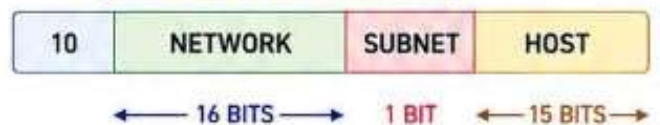
CLASS B – ONE BIT SUBNETTING

In Class B one bit subnetting, we borrow 1 bit from the Host portion to create 2 subnets.

CLASS B IP ADDRESS



BORROW 1 BIT FROM HOST



DEFAULT CLASS B

Default Subnet Mask : 255.255.0.0
Network Bits : 16
Host Bits : 16
Number of Networks : 1
Hosts per Network : 65,534

AFTER BORROWING 1 BIT

New Subnet Mask : 255.255.128.0
Network Bits : 17
Host Bits : 15
Number of Subnets : $2^2 = 2$
Hosts per Subnet : $2^{15} - 2 = 32,766$

SUBNET MASK IN BINARY

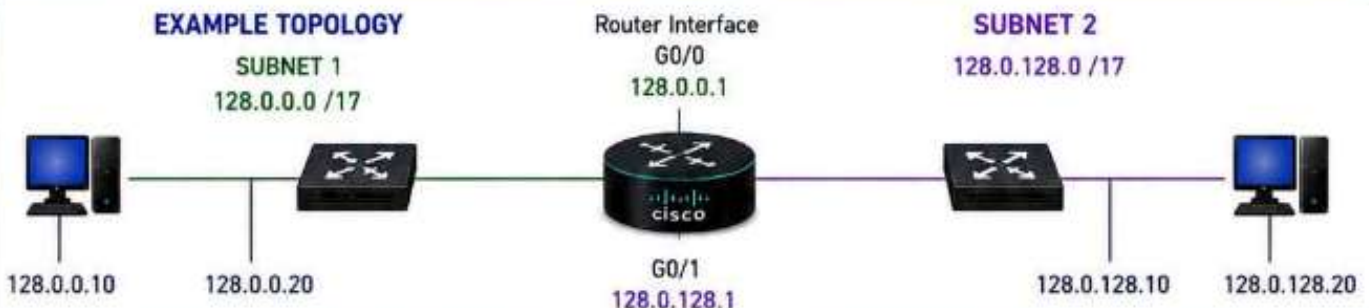
255.255.128.0
255 = 11111111
255 = 11111111
128 = 10000000
0 = 00000000

/17 (17 Network Bits)

SUBNETS CREATED

Subnet	Subnet ID	First Usable IP	Last Usable IP	Broadcast Address	Total Hosts
Subnet 1	128.0.0.0	128.0.0.1	128.0.127.254	128.0.127.255	32,766
Subnet 2	128.0.128.0	128.0.128.1	128.0.255.254	128.0.255.255	32,766

EXAMPLE TOPOLOGY



By borrowing 1 bit in Class B, we get 2 subnets with more efficient IP utilization while still providing a large number of hosts per subnet.

New Subnet Mask
255.255.128.0
(/17)

VLAN CONFIGURATION LAB

CREATE, ASSIGN AND VERIFY VLANs ON A SWITCH

OBJECTIVE

Configure VLANs on a switch, assign access ports to VLANs and verify connectivity between VLANs.

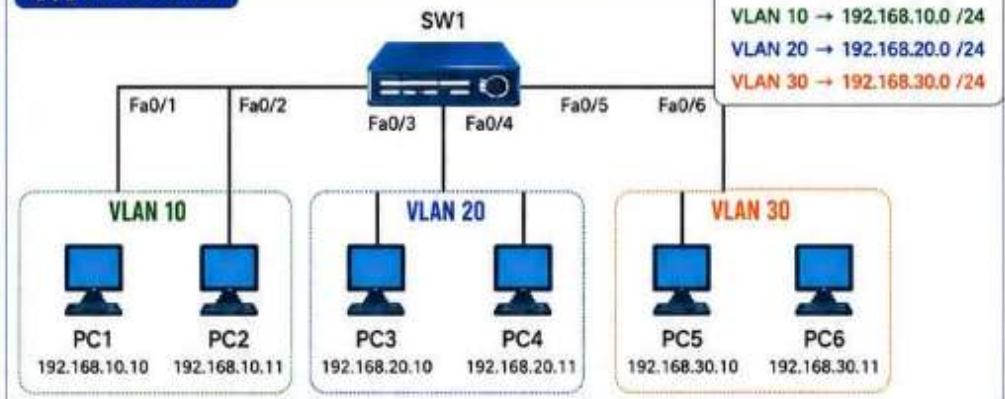


TASKS

- 1 Create VLANs 10, 20 and 30.
- 2 Assign ports to respective VLANs.
- 3 Configure access mode on ports.
- 4 Verify VLAN configuration.
- 5 Test connectivity within the same VLAN.
- 6 Verify no connectivity between different VLANs.



TOPOLOGY



PORT TO VLAN MAPPING

Port	Assigned VLAN	Mode	Description
Fa0/1	10	Access	PC1
Fa0/2	10	Access	PC2
Fa0/3	20	Access	PC3
Fa0/4	20	Access	PC4
Fa0/5	30	Access	PC5
Fa0/6	30	Access	PC6

VERIFICATION CHECKS

- ✓ VLANs are created.
- ✓ Ports are in correct VLANs.
- ✓ PCs in same VLAN can ping each other.
- ✓ PCs in different VLANs cannot ping each other.



SW1 - VLAN CONFIGURATION

```
1 enable
2 configure terminal
3 vlan 10
4 name SALES
5 vlan 20
6 name HR
7 vlan 30
8 name FINANCE
9 end
10 write memory
```



SW1 - ASSIGN PORTS TO VLANs

```
1 configure terminal
2 interface range fa0/1 - 2
3 switchport mode access
4 switchport access vlan 10
5 interface range fa0/3 - 4
6 switchport mode access
7 switchport access vlan 20
8 interface range fa0/5 - 6
9 switchport mode access
10 switchport access vlan 30
11 end
12 write memory
```



IMPORTANT SHOW COMMANDS

- show vlan brief
- show interfaces status
- show interfaces fa0/1 switchport
- show running-config



These commands help you verify VLANs, port status and assignments.

SAMPLE OUTPUT - SHOW VLAN BRIEF

VLAN	Name	Status	Ports
1	default	active	Fa0/7, Fa0/8, Fa0/9, Fa0/10, Fa0/11, Fa0/12, Fa0/13, Fa0/14, Fa0/15, Fa0/16
10	SALES	active	Fa0/1, Fa0/2
20	HR	active	Fa0/3, Fa0/4
30	FINANCE	active	Fa0/5, Fa0/6

TESTING CONNECTIVITY

Source PC	Destination PC	Expected Result
PC1 (VLAN 10)	PC2 (VLAN 10)	Ping Success
PC3 (VLAN 20)	PC4 (VLAN 20)	Ping Success
PC5 (VLAN 30)	PC6 (VLAN 30)	Ping Success
PC1 (VLAN 10)	PC3 (VLAN 20)	Ping Fail
PC2 (VLAN 10)	PC5 (VLAN 30)	Ping Fail
PC4 (VLAN 20)	PC6 (VLAN 30)	Ping Fail

CHALLENGE QUESTION

If you connect PC1 (VLAN 10) to PC3 (VLAN 20) and the ping is still failing even though both have correct IP addresses, what is the likely reason and how would you fix it?

Comment your answer below! 🗨️

QUICK TIPS



Always save configuration (write memory).



Use meaningful VLAN names.



Verify mode and access VLAN on ports.



Test connectivity after configuration.



Document your VLAN plan clearly.



ROUTER CONFIGURATION

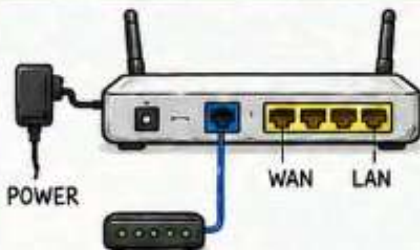
Step-by-Step Guide to Setup Your Router



★ WHAT YOU NEED

1. Modem
2. Router
3. UTP Cable (Ethernet Cable)
4. Computer / Phone

1. CONNECT THE HARDWARE



- Connect the modem to the WAN port.
- Connect your computer to a LAN port or connect via Wi-Fi.
- Connect the power adapter and turn on the router.

2. LOGIN TO ROUTER

192.168.1.1

Username: admin
Password: admin

Login

- Open a web browser.
- Type 192.168.1.1 in the address bar.
- Enter the default username and password (admin / admin).
- Click Login.

3. INTERNET (WAN) SETUP

Quick Setup
Network
WAN
LAN
Wireless
DHCP
System Tools

WAN Type: PPPoE

Username: your_isp_username
Password: your_isp_password

Save Next

- Select WAN Type (PPPoE / DHCP / Static).
- Enter your ISP Username and Password.
- Click Save and Next.

4. WI-FI (WIRELESS) SETUP

Quick Setup
Network
Wireless
- Wireless Settings
- Security
DHCP
System Tools

Wireless Name (SSID): My_WiFi

Security: WPA2-PSK

Password: 12345678

Save

- Enter a Wi-Fi Name (SSID).
- Select Security Type (WPA2-PSK).
- Enter a strong Wi-Fi Password.
- Click Save.

5. LAN SETTINGS

Quick Setup
Network
- WAN
- LAN
Wireless
DHCP
System Tools

IP Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Save

- The default LAN IP is 192.168.1.1
- Subnet Mask is usually 255.255.255.0
- Click Save.

6. REBOOT



Configuration Saved
Rebooting...

100%

- Your router will reboot.
- Reconnect to the Wi-Fi and enjoy the internet! 😊

💡 IMPORTANT TIPS

- ✓ Always change the default username and password.
- ✓ Use a strong Wi-Fi password.
- ✓ Keep your router firmware updated.
- ✓ Save your settings after any change.



SECURE

Change default settings to keep your network safe.



STABLE

Proper configuration ensures a stable connection.



FAST

Good settings improve internet speed.



RELIABLE

Backup & restore settings for future use.

LACP: HOW BANDWIDTH IS AGGREGATED

 LACP BALANCES FLOWS, NOT PACKETS




2 x 10 Gbps
= 20 Gbps
AGGREGATE

1 ONE FLOW

One TCP session



MAX THROUGHPUT
10 Gbps

2 MULTIPLE FLOWS

Multiple TCP sessions



AGGREGATE THROUGHPUT
20 Gbps

3 ONE HOST MULTIPLE CONNECTIONS

Multiple TCP sessions



POTENTIAL THROUGHPUT
UP TO **20 Gbps**



SINGLE FLOW = **10 Gbps** | MULTIPLE FLOWS = **UP TO 20 Gbps** AGGREGATE

STACK vs vPC

CATALYST STACK (STACKWISE)

ONE LOGICAL SWITCH



 SINGLE CONTROL PLANE

 ONE CONFIGURATION

 ONE MANAGEMENT IP

NEXUS vPC

TWO INDEPENDENT SWITCHES



 TWO CONTROL PLANES

 TWO CONFIGURATIONS

 TWO MANAGEMENT IPs

BANDWIDTH BEHAVIOR IS THE SAME. **ARCHITECTURE** IS DIFFERENT.